AIrtist: Agentic Concept Development for Contemporary-Art Installations

Pavel Serdyukov

Preprint, August 2026

Abstract. Early-stage contemporary-art installation development often begins from an underspecified artist-authored problem rather than a resolved artifact. We introduce AIrtist, an agentic system that develops such briefs into five complete project hypotheses through persistent proposal lineages, domain-structured review, contestable intervention, and separate terminal ranking. Each incumbent is reviewed into an opportunity state that distinguishes what should be preserved from what should change; Focused and Structural challengers then compete with the unchanged incumbent rather than automatically replacing it. We evaluate AIrtist on 50 held-out commission-style briefs using exhaustive blind 5×5 candidate-pool comparisons, exact A/B presentation reversal, and a separately instantiated GPT-5.6 Sol holistic pairwise evaluator. Against a substantially matched shared-parent generic iterative control, AIrtist yields a centered complete-pool effect of +0.0836 (95% CI [+0.0340, +0.1304], Holm $p = .0015$). Across 250 lineages, 149 final states remain incumbent-identical; among 101 adopted replacements, challengers receive mean exact-parent preference 0.693 and a brief-balanced centered effect of +0.437. Terminal selection is less resolved: Hybrid's +0.0533 Utility@1 advantage over Base is not statistically resolved. These results separate candidate-set construction, intervention quality, and attention allocation as distinct empirical problems. The evidence concerns textual proposals under model-mediated evaluation; it does not establish professional curatorial judgment, realized-artwork quality, or human co-creative benefit.

Keywords: contemporary art; installation art; concept development; creative agents; computational creativity; human–AI co-creativity; agentic AI.

1 Introduction

A consequential part of contemporary-art practice occurs before there is an artifact to fabricate, render, or exhibit. An artist may begin with a pressure rather than an image: a social relation that should become physically operative, a site whose existing use should not be erased, a material restriction, an institutional contradiction, or a conviction that several obvious artistic solutions would be wrong. At this stage, the artist may know what matters without yet knowing what the work should be. The problem is not how to execute a resolved concept, but how to develop alternative propositions for what the artwork could become.

Recent generative models make this stage newly accessible to computational support. They can produce plausible project descriptions, variations, references, images, and technical elaborations rapidly. Yet abundance of plausible completions does not by itself solve early concept development. In a contemporary-art installation, the same premise can lead to projects with radically different artistic operations: a concern may be represented symbolically, made causal through a material process, embedded in a spatial condition, delegated to an institutional procedure, or enacted through a spectator's position. A proposal can become clearer, longer, more interactive, or more technically sophisticated while becoming artistically weaker. Several incompatible directions may also remain viable at once. The computational problem is therefore not simply to generate more material, but to construct useful alternatives, decide what in an existing proposal should survive intervention, and recognize which resulting possibilities deserve attention.

Installation practice makes these difficulties particularly visible because the identity of a work can depend on relations among material, bodies, space, duration, site, and institutional framing rather than on a detachable artifact alone [1, 2]. We use artistic operation for the relation or mechanism through which a proposal enacts its stakes: what the work actually does through its material, spatial, temporal, spectator, or institutional organization. This does not define artistic quality, but it gives concept development a more discriminating computational object than topic, style, or visual appearance. A system working at this stage must reason not only about whether a proposal is coherent, but about which relation gives it identity and what kinds of change would preserve, sharpen, or destroy that relation.

Recent artistic practice already shows generative systems participating while the identity of a future artwork is still being articulated, including Kurant's Errorism [3], Lund's In the Middle of Nowhere [4], and Cai Guo-Qiang's cAI [5, 6]. Section 3 examines these precedents in detail.

These practices establish the use case but remain artist-specific procedures embedded in particular oeuvres and projects. Adjacent computational research, meanwhile, provides sophisticated machinery for early ideation, including professional concept design and scientific hypothesis generation, iterative review, and multi-candidate development [7, 8, 9]. What remains largely unformalized is the task between them: taking a previously unseen, under-specified contemporary-art problem and systematically developing several complete proposals for what the artwork itself could become.

We introduce AIrtist, an agentic system for this task. An artist supplies a rough contemporary-art installation brief describing the pressure they want to investigate, conditions that must remain consequential, directions they want to avoid, and—where relevant—site, material, institutional, or production constraints. We use concept development for this domain-facing task; computationally, AIrtist represents it as search over persistent proposal states. The system treats the brief as a provisional problem specification rather than as a description of a desired artifact, develops five complete project hypotheses, allows existing states to survive critique, and finally ranks the resulting five-proposal set.

To our knowledge, AIrtist is the first evaluated computational system whose primary task is to develop an under-specified artist-authored contemporary-art installation brief into multiple complete hypotheses for what the artwork itself could become. The computational object is the artwork proposition itself: an artistic operation together with its material, spatial, spectator, field, and institutional realization.

A held-out example illustrates the distinction. Brief 053 asks for an installation about contested maintenance using canvas, galvanized mesh, and hand-mixed pigment, with every component repairable in public during opening hours. One parent proposal, Echoing Gutter, already provides a coherent mechanism: pigment stains a mesh curtain, visitors brush the streak away, and the work returns toward a relatively uniform state. The proposal addresses damage and public repair, but its operation is completely reversible. Damage occurs, maintenance removes its trace, and the installation approaches its previous condition.

AIrtist's selected Structural challenger, Layered Ledger, retains the public act of brushing but places canvas behind the mesh. Cleaning the front now transfers residual pigment onto the backing. Repair therefore has two simultaneous consequences: one surface is restored while another accumulates the history of previous damage. The topic remains contested maintenance, but the artistic operation changes from damage $\rightarrow$repair $\rightarrow$reset to damage $\rightarrow$repair $\rightarrow${surface restored, history accumulated}. The useful intervention is neither a different illustration of maintenance nor additional technical detail. It changes what maintenance does.

This example also exposes why generic iterative refinement is an incomplete model of the task. A critique may correctly identify a weakness while the proposed repair destroys the strongest relation in the parent. Conversely, a proposal may contain a valuable operation even when some part of its realization should be replaced. Additional inference should therefore be able to test a possible transformation without making transformation mandatory. Likewise, producing several proposals creates a second problem: a strong direction may already exist in the candidate set while a ranking policy fails to surface it. Concept development, intervention, and selection are consequently related but empirically separable problems.

AIrtist operationalizes this view through five persistent proposal lineages. Each incumbent first receives heterogeneous specialized review. The resulting evidence is compiled into an opportunity state that identifies what should survive, the dominant liability, a familiar field logic to escape, an unused material or spatial opportunity, a decisive transformation to attempt, and additive fixes that should not count as progress. Two challengers are then generated independently from the same frozen parent. A Focused reconstruction rebuilds the diagnosed weak part while preserving the incumbent property marked for protection; a Structural escape is additionally permitted to replace the dominant operation or realization when staying closer would preserve the diagnosed problematic logic. Crucially, the incumbent remains eligible. Generating a challenger therefore does not require accepting it. After one state survives in each lineage, a separate terminal selector ranks the five resulting proposals.

The evaluated system is intentionally autonomous within this bounded stage: the artist provides the brief once and receives five ranked project hypotheses without intervening during the run. AIrtist is therefore a concept-development backend rather than a complete conversational co-creative partner. Negotiation over diagnoses, criteria, branches, and longer-term artistic memory is left to future artist-facing systems.

The experimental design follows the same decomposition. RQ1 asks whether domain-structured development produces a stronger five-proposal set than raw generation, a generic self-refinement pass, and a substantially matched generic iterative process. The strongest control, M5, starts from the exact same five self-refined parent proposals as AIrtist, maintains the same number of lineages and challengers, keeps the incumbent eligible, and shares challenger gating and lane comparison, while removing AIrtist's specialized review, opportunity representation, and differentiated intervention roles. RQ2 follows proposal identity through development and asks whether replacements adopted by the frozen policy are externally preferred to the exact parent states they displace. RQ3 holds the generated five-proposal set completely fixed and changes only the terminal ranking policy, isolating recognition from generation.

We evaluate the system on 50 held-out commission-style briefs. For RQ1, every AIrtist proposal is blindly compared with every proposal in each of three control pools, yielding complete 5 $\times$ 5 cross-pool matrices rather than evaluation of only system-selected winners. Each logical comparison is repeated under exact A/B presentation reversal. The primary external evaluator is a separately instantiated GPT-5.6 Sol holistic pairwise judge that does not observe system identity, lineage history, internal scores, or the external candidate utilities later derived from its judgments. Statistical inference is performed over briefs rather than individual candidate comparisons.

The results distinguish the three stages rather than producing a uniform end-to-end success story. For candidate-set construction, AIrtist significantly exceeds all three planned controls after Holm correction. Against the substantially matched M5 condition, the centered complete-pool effect is +0.0836 (95% CI [+0.0340, +0.1304], Holm $p = .0015$); secondary diagnostics indicate that the gain is not confined to one oracle-best candidate. For intervention, 149 of 250 final lineage states remain content-identical to their incumbents. Among the 101 replacements actually adopted, challenger preference against the exact displaced parent is 0.693, with a brief-balanced centered effect of +0.437 (95% CI [+0.266, +0.594]). Selection is less resolved. The Hybrid selector improves mean Utility@1 by +0.0533 over Base on identical candidate pools, but the confidence interval includes zero; it identifies an externally highest-utility candidate first on only 26% of briefs. The experiment therefore provides strong evidence for improved candidate-set construction and adopted intervention quality while identifying terminal recognition as a distinct unresolved problem.

These results suggest a broader diagnostic view of creative agents. A weak surfaced proposal can arise because the system failed to construct strong alternatives, because an intervention damaged an existing state, or because ranking hid useful material that had already been generated. Evaluating only a final top-ranked output collapses these failure modes. Contemporary-art concept development makes the distinction especially visible because a change in material causality, spatial organization, spectator position, or institutional procedure can alter the identity of the proposal rather than simply improve its presentation. Generative assistance can induce fixation [10] and, in other settings, improve individual outputs while narrowing collective diversity [11], reinforcing the need to distinguish candidate quality from the mere existence of multiple branches.

The paper makes four main contributions:

1. Contemporary-art concept development as an AI task. We formulate and evaluate early-stage installation concept development from under-specified artist-authored briefs, where the computational object is a set of alternative project hypotheses rather than the realization of an already resolved artwork.

2. A contestable agentic development architecture. AIrtist maintains persistent proposal lineages, converts heterogeneous review into an explicit opportunity state, generates Focused and Structural interventions from the same incumbent, and allows the incumbent to survive. The architecture represents what should remain invariant separately from what may need to change.

3. A multi-level evaluation methodology for open-ended creative agents. We separately evaluate the quality of the sampled possibility set, adopted interventions against their exact parents, and terminal ranking over identical generated candidate sets, including a substantially matched shared-parent iterative control.

4. Empirical evidence about where agentic value and error arise. Domain-structured development improves measured candidate-set quality over matched generic iteration; adopted replacements are preferred to the exact states they replace on average; and terminal selection remains an independent recognition problem even when stronger measured alternatives already exist in the generated set.

The claims remain bounded by the evaluation setting. AIrtist does not establish autonomous artistic agency, professional curatorial judgment, or improved human creativity. The evaluated objects are textual project hypotheses, the principal quality signal is model-mediated, and no artist negotiates with the system inside the frozen experimental pipeline. The present study instead asks whether early contemporary-art concept development can be made into an explicit computational task and whether the proposed agentic development policy produces measurable improvements under a disclosed evaluation procedure.

Paper organization. Section 2 develops the contemporary-art formulation underlying the task, including artistic operation and the bounded working quality model. Section 3 positions AIrtist relative to contemporary artistic practice, AI-assisted ideation, and agentic refinement. Section 4 describes the AIrtist architecture, and Section 5 specifies the benchmark, controls, external evaluation, and inference procedure. Sections 6 and 7 report the main results and mechanism/failure analysis. Section 8 discusses implications for creative-agent design; Section 9 develops the intended artist-facing use and future negotiated co-creativity; Section 10 states limitations and threats to validity; and Section 11 concludes.

2 Representing Contemporary-Art Installation Ideation as a Computational Problem

AIrtist does not generate a physical installation. It generates a proposal for a future installation: a textual account of what would be installed, how its material and spatial organization would behave, how spectators would encounter it, what institutional conditions it would depend on, and why those choices belong to the artistic proposition rather than merely to its fabrication. The computational object in this paper is therefore not the realized artwork but a sufficiently developed installation hypothesis. The dossier is intended to be sufficiently specified for proposal development and comparative judgment; it is not assumed to be sufficiently specified for fabrication without further artistic, curatorial, engineering, conservation, and site-specific decisions. A realized installation may depart substantially from its dossier through scale, fabrication, material accident, maintenance, institutional negotiation, audience behavior, or later artistic decisions.

This distinction matters because a language model can readily generate a title, a room, a set of objects, a material palette, and persuasive curatorial prose. None of these, separately or together, guarantees that the text specifies a coherent installation. Two dossiers may address the same topic and use almost the same materials while proposing fundamentally different works because bodies move differently through them, matter changes differently over time, access is withheld under different conditions, or the institution plays a different role. Conversely, works that look unrelated at the level of object vocabulary may depend on closely related structures of delay, depletion, redistribution, visibility, passage, or institutional control.

The study therefore makes a limited representational assumption: A textual dossier can describe a proposed installation in enough detail to support meaningful computational reasoning about the artistic situation it is intended to produce. "Enough detail" here has a practical meaning. The proposal should state what exists, what changes, what depends on what, what a spectator can perceive or do, which material properties matter, and which production or institutional conditions are necessary. It need not—and in practice cannot—determine every property of the eventual artwork.

For AIrtist, the first problem is consequently not generation but representation. Fluent proposal prose easily collapses three different questions:

1. What artistic problem, tension, or pressure is the proposal trying to address?
2. What actually happens in the proposed installation that gives that problem a specific artistic form?
3. What concrete spatial, material, temporal, perceptual, technical, and institutional arrange-

ment makes that happening possible?

We refer to these levels as artistic stakes, artistic operation, and realization. The section develops this representation in two stages. First, installation theory and established artistic practice motivate why space, duration, material behavior, embodied spectatorship, site, and institutional context may be constitutive rather than incidental. Second, we translate those concerns into an explicit, bounded computational surrogate of proposal quality for one task: developing contemporary-art installation proposals under museum-commission conditions. The theoretical literature motivates what cannot safely be omitted from the representation; it does not directly entail our ten computational dimensions, their decomposition, or their use inside AIrtist.

2.1 Installation as a Situated Configuration

A useful starting point is that installation is rarely exhausted by a self-contained object. Bishop foregrounds the spectator's bodily presence within the work; Petersen describes installation through activated physical space and context, temporal extension, and embodied perception; Kwon traces practices in which site becomes physical, institutional, social, or discursive rather than merely locational; O'Doherty makes the complementary argument that the gallery is not a neutral container; and Rebentisch places the unstable relation among work, site, spectator, autonomy, and public context at the center of installation aesthetics [1, 12, 2, 13, 14]. These positions are not interchangeable, and AIrtist does not collapse them into one definition of installation. They motivate a narrower premise useful for computation:

Working premise. An installation may be constituted not only by the intrinsic properties of objects but by specific dependencies among material, space, duration, spectator, and institutional context. The historical literature makes the move away from the isolated object unusually explicit. Petersen writes that installation "activates the physical space and the contexts in which it is embedded" and emphasizes its temporal and embodied character [12]. Morris, writing from the Minimalist debates of the 1960s, argues that decisions internal to the object can no longer be separated cleanly from decisions external to its physical presence: placement, scale, surrounding space, light, and the viewer's changing position become part of the conditions under which the work is experienced [15]. Burnham's systems aesthetics makes an even broader relational claim: "art does not reside in material entities, but in relations between people and between people and the components of their environment" [16].

We do not adopt any of these statements as a ready-made computational definition. Their relevance is practical. They warn against representing an installation as a list of objects plus a theme. If changing a route, a temporal condition, a spectator position, a material behavior, or an institutional rule can change what the work does, those elements must be available to the computational system as first-class parts of the proposal.

Table 1 makes this intuition concrete across heterogeneous practices. The second column supplies the physical description necessary to understand the analytical reading; the third and fourth columns are our interpretation rather than terminology attributed to the artists. The examples are deliberately incompatible in medium, politics, temporality, spectator agency, and visual form. Their common value for our problem is partly negative: they show what a computational representation should not assume. An installation need not be interactive; material transformation need not be spectacular; the spectator need not make a choice; a bounded object need not be the principal unit of the work; and institutional procedure can be as constitutive as fabricated matter. AIrtist therefore allows a proposal's organizing logic to arise through perception, material behavior, image, object presence, atmosphere, spatial organization, duration, spectatorship, social relation, procedure, or institutional condition. No one of these receives an automatic quality bonus. The system contract explicitly guards against equating contemporary art with participation, technology, procedurality, or institutional critique at the expense of stillness, image, sound, atmosphere, object presence, fiction, or ambiguity.

2.2 Artistic Operation

The central abstraction used by AIrtist is artistic operation. The term does not mean an algorithm, a motorized mechanism, or an instruction that a visitor must execute. We use it for the main action, dependency, restriction, sequence, material change, access condition, perceptual condition, or spatial procedure through which the installed work makes its artistic problem happen rather than merely describing it.

In plainer language, an artistic operation answers: What actually happens because this installation has been arranged in this particular way? What changes, for whom or for what,

Table 1: Operational readings across heterogeneous installation practices.

Work	What a visitor physically encounters	AIrtist-style analytical reading of what happens	Conditions that carry it
James Turrell, *A Frontal Passage* (1994)	A visitor enters a dark chamber divided by an apparently sharply bounded field of red light. The source is hidden; the light appears to end in space as if it had volume.	Viewer position and visual adaptation ↔ controlled light field: perception converts immaterial light into an apparently solid presence.	Darkness, hidden light source, architectural enclosure, viewing position, adaptation over time [17].
Mel Bochner, *Measurement Room* (1969)	Tape lines and numerical measurements are applied directly to walls, openings, windows, and passageways of the room itself.	Existing architecture ↔ explicit measurement of that architecture: the room ceases to function merely as a container and becomes perceptible as a measured system.	The actual room, its dimensions, tape and notation, the viewer inhabiting the measured space [18].
Hans Haacke, *Condensation Cube* (1963–67)	Water sealed in a transparent acrylic cube evaporates and condenses; droplets form and move differently as surrounding conditions change.	Water system ↔ ambient climate: environmental conditions continually produce the visible state of the work.	Water, transparent enclosure, temperature differential, humidity, duration, local environment [19].
Doris Salcedo, *Shibboleth* (2007)	A long fissure interrupts the otherwise continuous floor of Tate Modern's Turbine Hall and the route across it.	Institutional architecture and passage ↔ physical division: a surface normally experienced as continuous becomes structurally discontinuous and bodily negotiated.	The museum floor, fissure, scale of the hall, visitor movement and avoidance [20].
Felix Gonzalez-Torres, "Untitled" (Portrait of Ross in L.A.) (1991)	A pile begins at an ideal weight of 175 lb. Visitors may take individually wrapped candies; the pile diminishes and may later be replenished.	Visitor removal ↔ changing material mass ↔ institutional replenishment: loss and restoration become repeated material events rather than represented images.	Removable candy, ideal weight, permission to take, variable pile size, replenishment protocol [21].
Tania Bruguera, *Tatlin's Whisper #5* (2008)	Mounted police enter the museum and use crowd-control techniques to direct, divide, and corral visitors.	Police crowd-control procedure ↔ visitor movement: spectators do not merely view represented authority; their own bodies become subject to its procedures.	Trained mounted police, museum public, crowd-control procedures, duration, institutional permission and risk [22].
Rirkrit Tiravanija, *untitled 1992 (free)*	Gallery space becomes a functioning place for cooking and serving free Thai curry; visitors eat, remain, and interact there.	Gallery institution ↔ hospitality activity ↔ visitors sharing time: an ordinary social practice reorganizes what exhibition space is being used for.	Kitchen infrastructure, food preparation, service, duration, visitors, institutional hosting [23].
Fred Wilson, *Mining the Museum* (1992–93)	Objects from an existing historical collection are regrouped, juxtaposed, labeled, and displayed under altered curatorial arrangements.	Collection objects ↔ museum classification and display conventions: curatorial organization becomes the means through which suppressed histories become perceptible.	Existing collection, vitrines, labels, grouping, museum categories and institutional history [24].

and what does that change make perceptible or experienceable? Whenever we describe an operation as a relation, the terms of that relation should therefore be identifiable.

In Turrell, the relevant dependency is between a situated viewer and a controlled field of light: under particular viewing conditions, light appears to acquire a boundary and presence that it does not possess as an ordinary illuminated surface [17]. In Haacke, it is between water inside a transparent enclosure and the surrounding climate: temperature and humidity alter evaporation and condensation, and therefore alter what the spectator sees [19]. In Gonzalez-Torres, it is among visitor removal, the diminishing mass of the candy pile, and the institution's possibility of replenishment [21]. In Bruguera, it is between police crowd-control procedures and the movement of museum visitors [22].

The word consequential occasionally used below has this limited meaning. It does not mean socially important or morally consequential. A dependency is consequential for the proposed work when changing that dependency changes what the installation permits, withholds, transforms, exposes, accumulates, erodes, redistributes, or makes perceptible. Thus, rather than saying only that "the spectator relation is consequential," we should be able to say what the relation is and what follows from it: visitors remove candies → the pile loses mass; humidity changes → condensation changes; police maneuver → visitors' routes change; the viewer moves relative to the light field → its apparent spatial character changes.

A relation that can disappear while leaving the proposal functionally unchanged is probably descriptive context rather than part of the dominant operation.

Similarly, installed situation simply means the configuration that would exist once the proposal were realized: objects and materials, architecture, duration, active processes, spectators, and relevant institutional rules considered together. The term is broader than "installation object" because the decisive condition may exist between these elements rather than inside one of them.

Artistic operation is AIrtist's computational term, not an established art-historical category. It gives the system a level of description between a theme—too abstract to guide revision—and a realization—too specific to tell us what should survive when particular materials or mechanisms

Table 2: Stakes, operation, and realization as an analytical scaffold.

Work	Possible articulation of the stakes	Operation: the specified dependency	Realization
Turrell, *A Frontal Passage*	Instability of the boundary between immaterial perception and apparently physical presence.	Situated viewer ↔ controlled red-light field: under specified viewing conditions, light is experienced as having a palpable boundary.	Dark approach and chamber, hidden light source, sharply controlled red field, viewing trajectory [17].
Bochner, *Measurement Room*	How abstract measurement systems organize the way physical space is understood.	Actual architecture ↔ numerical description of that same architecture: measurement is superimposed on the room it measures.	Tape and numerical notation applied directly to walls, doors, windows, and passages of the specific room [18].
Haacke, *Condensation Cube*	A work as an ongoing physical system rather than a stable represented image.	Sealed water ↔ changing environmental conditions: temperature and humidity continually produce different condensation states.	Transparent acrylic enclosure, water, ambient climate, time [19].
Gonzalez-Torres, "Untitled" (Portrait of Ross in L.A.)	Loss, bodily vulnerability, love, disappearance, and the possibility of renewal in relation to Ross Laycock and AIDS.	Visitor removal ↔ diminishing mass ↔ optional institutional replenishment: loss is enacted as a material process that can be repeatedly renewed.	175-lb ideal weight, individually wrapped candies, visitor permission to take them, variable installation, replenishment [21].
Bruguera, *Tatlin's Whisper #5*	Authority, coercion, and the bodily reality of political control.	Police crowd-control procedure ↔ museum visitor movement: control acts on the spectator's own body and route.	Mounted officers, museum visitors, trained maneuvers, institutional permission, limited duration [22].

change.

2.2.1 Stakes, operation, and realization

For proposal development, we separate three analytical questions:

- Artistic stakes: What tension, contradiction, experience, or question makes the proposed work worth pursuing?
- Artistic operation: Which specified dependency makes that tension occur in the proposed work rather than merely being named by it?
- Realization: What concrete materials, spatial arrangement, timing, technical system, and institutional conditions instantiate that operation here?

These are not claimed to be ontological components present in every artwork. They are a computational scaffold for search: three questions a system must answer separately if it is to alter one part of a proposal without accidentally erasing another.

The distinction becomes clearer when applied to existing works. The same topic could be realized through a different operation, and similar operations could support different stakes. That is why AIrtist keeps the levels separate. "Loss" is not an operation. Neither is "candy." "Visitors remove material from a body-like total, diminishing it, while the institution may restore that total" is closer to an operation because the participating elements and their dependency are explicit.

2.2.2 What counterfactual substitution tells us

Counterfactual questions help identify whether a dependency is constitutive, but they should not be mistaken for a philosophical test of artwork identity. Consider the Gonzalez-Torres example [21]. Three superficially similar installations would establish different operative conditions.

A permanent resin sculpture visually imitating the candy pile. The approximate image and initial mass could be preserved, but the dependency visitor taking $\rightarrow$ diminishing material body would disappear. The object could still refer to loss, but it would no longer enact loss through repeated public depletion. Real candies that visitors may take, with no possibility of replenishment. The dependency between visitor removal and diminishing mass would remain, but it would become one-way. The temporal structure would be irreversible depletion toward disappearance.

Real candies that visitors may take and the institution may replenish. Removal and replenishment form a repeated dependency among spectators, material mass, and institutional care. Loss remains materially present, but so does the possibility of restoration. These examples are not meant to assert that changing one condition necessarily produces an entirely different artwork or one determinate new meaning. The diagnostic question is narrower:

Which claimed artistic effects depend on which concrete dependency? If a dependency can be removed without changing what the proposal asks the spectator, material, space, or institution to do, AIrtist should not treat it as central to the operation. This also clarifies what we mean by preserving an operation. Preservation does not require literal conservation of every object or device. It requires preserving the dependency judged central to the proposal.

2.2.3 Operation is not mechanism

Large language models are particularly fluent at jumping from topic to device. Memory becomes an archive; surveillance becomes a camera; inequality becomes uneven platforms; absence becomes an empty chair; climate becomes fog, heat, or a live data display. Curatorial prose can then make the device sound conceptually inevitable after the fact. AIrtist therefore distinguishes a mechanism from the artistic operation it serves. A camera, motor, pump, pressure sensor, projector, pile of bricks, speaker, curtain, or computer is a component of realization. It becomes artistically necessary only through what dependency it establishes.

For the public-shame example used later in the paper: Level Example Stakes Public shame and the withholding of social recognition. Mechanism Infrared camera, video buffer, projector, translucent scrim. Operation A passerby encounters a public presence only after the people associated with that presence are absent. Realization Visitor movement is captured, delayed, and projected as silhouettes onto a scrim across the route.

If cameras become inadmissible, AIrtist can ask whether the delayed relation can survive another realization. This is exactly what occurs when Delayed Projection is challenged by Absent Passage: recording is discarded while temporal displacement remains available for development.

2.2.4 Operation is not participation

Operation should also not be confused with audience activity. Turrell establishes an unusually specific relation to a spectator who need not press, contribute, disclose, or collaborate. Bruguera makes visitor movement subject to an imposed procedure. Tiravanija makes shared activity constitutive. These are different spectator relations, not points on a scale from passive-and-weak to interactive-and-strong [17, 22, 23].

This distinction matters computationally because generative systems frequently answer criticism about "engagement" by adding buttons, sensors, choices, visitor submissions, or responsive media. AIrtist gives no automatic bonus for any of these moves. Its spectator criterion asks instead: Where is the spectator in the work's logic? What can or cannot happen from that position? What changes because the spectator is there?

This is why the term and the literal description should often appear together. "Spectator specificity" is useful shorthand, but its operational content is concrete: the spectator may have to stand at a particular distance, wait through a delay, cross a boundary, remove matter, be redirected, remain unable to act, or simply occupy a position from which a perceptual condition emerges. Specificity therefore does not require participation.

This refusal of a simple passive/active hierarchy has clear precedents. Ranciere questions the opposition that equates spectatorship with passivity, while Bishop argues against treating participation as an automatic aesthetic or political virtue [25, 26]. These positions do not define AIrtist's spectator metric, but they support its refusal to equate activity with quality.

2.2.5 Why AIrtist asks for one dominant operation

AIrtist does in fact impose a stronger representational constraint here than a purely descriptive theory of installation would require. The proposal contract asks every candidate to articulate one clear dominant artistic operation, while allowing a supporting operation when it serves rather than competes with the dominant one. The generation and review prompts likewise ask whether a proposal is governed by one necessary operation rather than by a bundle of unrelated moves. This is therefore not merely terminology introduced later for explaining the system. It is part of AIrtist's search representation.

The constraint should nevertheless not be confused with an ontological claim about artworks: AIrtist does not assume that a realized artwork objectively possesses one true operation. It requires a proposal to articulate one dominant operation so that the current computational search has a stable object to preserve, challenge, or replace.

A realized installation may sustain several simultaneous, historically unstable, or mutually irreducible relations. But a computational revision process faces a narrower practical question: What, if destroyed by the next revision, would make this proposal cease to preserve the artistic logic we were trying to develop? Without a conservation target, every reviewer can motivate a different addition: stronger visual identity, clearer explanation, more participation, richer materiality, better accessibility, more explicit context, more technology. The dossier can become more developed as prose while no artistic dependency remains stable enough to judge whether the proposal has improved. The dominant operation therefore functions as an information bottleneck and conservation target for search.

It is also an economy constraint. AIrtist explicitly treats a simpler, more necessary dominant operation as preferable to an elaborate proposal whose multiple dependencies are not artistically required. A supporting operation is permitted when causally necessary; "one dominant operation" does not mean "one object," "one effect," or "one interpretation."

2.2.6 Counterfactual debugging without material essentialism

Three questions help AIrtist test whether a proposal has specified constitutive dependencies rather than decorative details:

1. Substitution. If this material, mechanism, or medium were replaced, which claimed artistic effect would disappear, and why?
2. Relocation. If the same components were installed so that route, threshold, distance, access, orientation, or duration no longer mattered, what would cease to happen?
3. Spectator reassignment. If the spectator could occupy any position, act or not act arbitrarily, and spend any amount of time without changing the proposal, is the stated spectator relation actually specific?

These are debugging heuristics, not necessary conditions for strong art. A work can deliberately permit variable materials, delegated fabrication, multiple sites, or different spectator positions. What matters is that the proposal distinguish what is variable from what must remain invariant for the operation to continue functioning.

Bochner's Measurement Room, for example, can be reinstalled in another gallery. Its exact dimensions are not invariant across realizations. What must remain is the dependency between the architecture actually present and the measurements applied to that architecture. Copying the numerical values from a previous room onto a new room would preserve the graphic style while breaking the operation [18].

Likewise, the exact arrangement of Gonzalez-Torres's candy pile may vary, but removability, ideal weight, and the possibility of replenishment are not equivalent to arbitrary visual styling [21]. Haacke's cube can exist under different climatic conditions precisely because environmental variation is part of the system. Replacing actual condensation with a prerecorded animation could preserve a recognizable image while removing the material dependency that produces it [19].

This is the sense in which AIrtist uses material necessity without assuming material essentialism. Bruno's account of materiality is useful here because it approaches surfaces not simply as substances but as sites of mediation among bodies, media, architecture, and perception [27]. The relevant computational question is therefore often not: Must it literally be this exact substance? but: Which property, behavior, history, interface, or institutional relation of the realization must remain active for the proposed operation to survive?

2.3 A Bounded Working Model of Proposal Quality

The argument so far concerns what an installation proposal needs to represent. It does not yet tell AIrtist how to search for a stronger proposal. Installation theory motivates attention to situated space, material behavior, duration, spectatorship, and context. It does not imply AIrtist's ten computational dimensions, their decomposition, their use in ranking, or the precise search pressures encoded by the system. Those are methodological choices. AIrtist therefore operates under an explicit bounded surrogate of installation-proposal quality. It is a computational and curatorial hypothesis about properties that make the class of museum-installation proposals studied here worth developing.

Computational-creativity research provides a methodological precedent for making such commitments explicit. Ritchie and Jordanous argue, in different ways, that creative-system evaluation should state what forms of value are being assessed rather than treating "creativity" or "quality" as undefined universal properties [28, 29].

Internally, AIrtist expresses the surrogate at two related levels. At the proposal-contract level, it uses broad quality principles concerning brief-specific stakes, artistic-operation necessity, visual-spatial identity, field-aware distinctiveness, and production/institutional viability. For detailed quality-bearing comparison, these concerns are operationalized as ten dimensions:

1. raw-prompt fidelity;
2. core-tension enactment;
3. artistic-operation necessity;
4. field distinctiveness;
5. spatial encounter;
6. visual encounter;
7. material necessity;
8. spectator specificity;
9. public-reception resilience;
10. production/institutional viability.

These dimensions are used as pressure tests and scoring constructs inside AIrtist; proposal generators are explicitly instructed not to self-score or recite them as curatorial rhetoric.

Table 3 groups the ten dimensions into five commitments to make their motivation easier to follow. The remainder of this subsection explains all ten dimensions.

Table 3: Five working commitments and ten computational dimensions.

Working commitment	Central question	Computational dimensions
Brief-specific artistic problem	Does the proposal address the actual commission pressure rather than merely reuse its vocabulary?	Raw-prompt fidelity; core-tension enactment
Necessary artistic operation	Does something specific happen through the work, and do its materials actually carry that happening?	Artistic-operation necessity; material necessity
Situated encounter	Is there a concrete perceptual, spatial, temporal, and spectator situation rather than only a concept description?	Spatial encounter; visual encounter; spectator specificity
Field-aware artistic proposition	Does the work transform rather than simply repeat a familiar operating format, and can it resist being exhausted by a generic reductive reading?	Field distinctiveness; public-reception resilience
Commission hypothesis	Could the described artistic relation plausibly exist under the stated institutional and production conditions?	Production/institutional viability

2.3.1 Raw-prompt fidelity and core-tension enactment

These dimensions address a common language-model failure: producing a fluent work near the topic while escaping the actual problem. Raw-prompt fidelity asks whether the proposal respects what the artist asked for: the site, materials, must-address conditions, must-avoid conditions, and concrete commission context. It is deliberately literal where the brief is literal. A proposal that uses cameras when cameras are prohibited should not recover merely because the resulting installation sounds sophisticated.

Core-tension enactment asks a different question. A proposal can satisfy every keyword while still avoiding the reason those constraints were introduced. A brief about unequal waiting, for example, can mention delay, clocks, and inequality while constructing nothing more specific than a visual metaphor for time. Enactment asks whether inequality in waiting changes something in the proposed situation—who waits, for how long, under what fixed condition, and what becomes perceptible because those durations differ.

Existing installations help clarify the distinction even though they were not produced from our benchmark briefs. Salcedo's Shibboleth is not simply a work that refers verbally or iconographically to division. Division enters the continuity of the Turbine Hall floor and therefore the visitor's actual route [20]. If the same political language were attached to an unaffected gallery floor, the topic might remain while the enacted condition would change radically.

The computational distinction is therefore:

- fidelity: did the proposal stay inside the problem the artist gave us?
- enactment: did that problem become structurally present in what the installation does?

2.3.2 Artistic-operation necessity and material necessity

Artistic-operation necessity asks whether the proposal depends on a specific organizing dependency rather than on explanatory prose, a generic mechanism, a decorative effect, or a familiar metaphor. Put literally: If the curatorial paragraph disappeared, what would the installation itself still do that carried the proposition?

Material necessity asks whether a chosen material or realization system contributes through a property, behavior, process, history, or institutional relation rather than functioning as an interchangeable visual sign. Haacke's Condensation Cube makes the connection unusually clear. Acrylic, water, and ambient climate do not merely symbolize change. Their physical relation continually produces the visible state of the work [19]. A video of condensation could resemble that state, but the environmental process itself would have disappeared. Gonzalez-Torres offers a different form of necessity. Candy is not important merely because it looks colorful or symbolizes sweetness. It can be individually taken, it forms a measurable and diminishing mass, and it can be replenished [21]. Those properties enable the visitor–material–institution dependency through which depletion and renewal take place. Morris's account of sculpture provides an earlier historical motivation for refusing to treat physical decisions as isolated formal properties. His argument that decisions concerning the object cannot be understood independently from its placement and surrounding conditions helps motivate a model in which material and spatial choices are judged through what they do in a situation rather than through named medium alone.

Material necessity therefore does not mean: If another material could ever be used, the proposal is weak. It means: If the material changes, the constitutive property or dependency should survive rather than only the visual appearance.

2.3.3 Spatial encounter, visual encounter, and spectator specificity

These three dimensions deliberately separate aspects that generative prose often bundles together as "immersive," "experiential," or "engaging." Spatial encounter asks how position, route, distance, access, threshold, scale, orientation, visibility, and duration organize the work. Put literally: Where does the visitor stand or move, what changes along that route, and why would the same objects arranged differently no longer produce the same work? Bochner's Measurement Room is a direct example. The room is both the measured thing and the space the spectator inhabits [18]. Spatial arrangement is not scenery around an autonomous object. Salcedo provides another. In Shibboleth, the floor across which a visitor would ordinarily move continuously becomes physically divided [20]. Movement in relation to the fissure is part of the encounter rather than a neutral viewing condition.

Visual encounter asks what can actually be perceived before the explanatory concept is supplied. This is not a score for beauty, spectacle, photogenic quality, or visual complexity. It asks whether the proposal produces a specific perceptual situation whose visible or sensible identity follows from its operation. Turrell is useful precisely because the work can be visually forceful without accumulating objects or spectacle. The apparent solidity of red light arises from the controlled dependency among light, darkness, chamber, and viewing position [17].

Spectator specificity asks who the spectator is in the logic of the work. Where are they? What do they know? What can or cannot happen from that position? Must they cross something, wait, remove something, be redirected, remain unable to act, look from one particular position, or experience a process over time? Thus the technical term and the literal account should be read together: spectator specificity = a proposal's explicit account of what position, attention, movement, action or non-action, access, duration, visibility, knowledge, or consequence is assigned to the spectator. Specificity does not require participation. Turrell's spectator looks and moves. Bruguera's spectator is subjected to direction. Tiravanija's visitor may eat and remain. Gonzalez-Torres's visitor may remove material [17, 22, 23, 21]. What AIrtist rejects is the generic statement that "the viewer interacts" or "the viewer reflects" without specifying what the interaction or reflection is materially contingent upon.

2.3.4 Field distinctiveness and public-reception resilience

Field distinctiveness is not novelty maximization. Contemporary art is historically and institutionally relational: a gesture derives part of its force from what kinds of gestures are already available, recognizable, exhausted, contested, or transformed. Bourdieu's field theory and

Danto's artworld argument provide different theoretical accounts of this relationality; Kwon demonstrates how site-specific and institutional practices make it artistically explicit [30, 31, 2]. AIrtist therefore asks not: Has anyone ever used this object before? but: If this proposal uses a recognizable artistic format, what changes in that format's operating logic here?

Wilson's Mining the Museum is useful because none of its basic display technologies—vitrines, labels, historical objects, taxonomic grouping—is novel. Its distinction lies in operating on the museum's own classificatory system and reorganizing what that system makes visible [24].

There is, however, an important epistemic limit to this criterion. AIrtist does not possess an authoritative map of contemporary art. Its field-context stage uses model-mediated recall of works, artists, formats, criticism, and neighboring contexts, not provenance-bearing bibliographic retrieval. Field evidence is therefore a working hypothesis about nearby artistic lines, useful for directing search but not sufficient to establish an art-historical claim that a proposal is objectively unprecedented or distinct. In this paper, field distinctiveness should accordingly be understood as field-aware differentiation under the system's recalled context, not as measured global novelty in contemporary art.

Public-reception resilience addresses a related but different failure. A proposal can contain a sophisticated concept internally while its public form collapses into a much flatter genre: interactive gadget, wellness environment, awareness display, memorial cliche, educational visualization, selfie backdrop, corporate engagement, or generic participatory service. The criterion does not demand opacity or antagonism. Nor does it penalize popular, therapeutic, spectacular, civic, or didactic work as categories. It asks: Does one obvious generic reading exhaust what the installation actually does, or does the installed relation continue to produce something that the label leaves unexplained?

Tiravanija illustrates why surface format is insufficient. "Serving curry in a gallery" can be redescribed as hospitality or catering, but the artistic proposition involves how exhibition space, authorship, social time, spectatorship, and institutional behavior are reorganized around that act [23]. Bruguera offers the complementary case. Mounted police are visually spectacular, but the work is not exhausted by the image of police inside a museum. Its specificity lies in actual crowd-control techniques changing the movement of the spectators themselves [22]. Remove that dependency and retain only the spectacle, and the work operates differently. Eco's account of openness is relevant because reception resilience is not the elimination of interpretive plurality. Ngai's analysis of the gimmick is useful for the opposite danger: a work can become legible through one immediately available reductive judgment that consumes the rest of its structure [32, 33]. AIrtist turns this concern into a bounded curatorial diagnostic rather than a universal theory of reception.

2.3.5 Production and institutional viability

AIrtist evaluates production/institutional viability because its computational object is a commission proposal for a future work. This should not be confused with the proposition that feasible art is better art. Deliberate impossibility, waste, fragility, institutional conflict, or excessive labor can be artistically meaningful. The weakness arises when the dossier depends on a contradiction it does not recognize: a material cannot behave as described; the required circulation makes the museum unsafe; the work assumes unlimited staff while treating that labor as invisible; a maintenance procedure silently changes the artwork; or the central effect depends on permissions the proposal could not plausibly obtain. Becker's account of art worlds is useful because it treats artistic production as collective activity supported by conventions, specialized labor, and material networks rather than by the isolated artist alone [34]. Jackson goes further for contemporary performance and socially engaged practice by showing how support, maintenance, administration, labor, and infrastructure can become artistically constitutive rather than merely backstage logistics [35].

The works in Table 1 make this visible in different ways. Gonzalez-Torres's replenishment is not simply a shipping detail; the exhibitor's possibility of replenishing the pile conditions the work's temporal behavior [21]. Tiravanija's kitchen, food service, sanitation, duration, and institutional hosting are inseparable from the proposed social situation [23]. Bruguera's mounted-police work depends on institutional permission, trained performers, public safety, and ethical risk [22]. In each case, changing the production structure can change the artistic relation itself.

The criterion therefore asks: Could this proposed dependency plausibly be produced, maintained, and encountered under the stated commission conditions—and if its difficulty is intentional, has the proposal made that difficulty part of the work rather than ignored it? Narrow

Table 4: Theoretical motivation and AIrtist-specific operationalization.

AIrtist construct	Closest theoretical/practical motivation	Status and boundary inside AIrtist
Situated installation configuration	Bishop; Petersen; Kwon; Morris; Rebentisch	Direct motivation for representing space, time, embodiment, site, and institutional context as potentially constitutive.
Artistic stakes	Broadly continuous with artistic problematics rather than one specific theoretical term	AIrtist analytical shorthand for the tension or question a proposal is organized around.
Artistic operation	Petersen's performative/modus-operandi account; Burnham's relational systems; Morris's expanded situation	AIrtist-specific abstraction for an explicit dependency through which the work does something rather than merely referring to a topic. The proposal contract requires one dominant operation for computational tractability.
Material necessity	Morris; process/system practices; Bruno	Diagnostic of dependence on material property, behavior, history, or interface; not material essentialism.
Spatial / visual encounter	Bishop; Petersen; Morris; installation and phenomenological discourse	Task-specific decomposition of the situated perceptual-spatial event.
Spectator specificity	Bishop; Petersen; Ranciere; participatory-art debates	Specificity of spectator position and consequence; activity itself receives no bonus.
Field distinctiveness	Bourdieu; Danto; Kwon; institutional critique	Field-aware transformation under model-mediated recalled context; not verified global novelty.
Public-reception resilience	Eco; reception debates; Ngai	Curatorial heuristic against reductive flattening; not a theory that difficult or opaque art is superior.
Production/institutional viability	Becker; Jackson; institutional and maintenance practices	Relevant because the object is a museum-commission hypothesis; feasibility is not treated as an independent aesthetic virtue.
One dominant operation	No art-theoretical equivalence required	Explicit representation and economy constraint used by generation and review; not an ontological claim that realized artworks objectively contain one true operation.
Holistic external preference	Comparative LLM evaluation	Held-out outcome measure that deliberately does not reproduce the ten-dimensional internal scoring decomposition.

eligibility and hard-safety gating remain conceptually separate from this quality judgment. AIrtist does not treat the hard gate as an eleventh aesthetic score: gating removes proposals with concrete ineligibility, hard-constraint, duplication, or severe safety failures before relative artistic comparison.

What the model deliberately does not reward Several properties that language models can amplify cheaply receive no independent quality bonus:

- novelty for its own sake;
- participation for its own sake;
- technological sophistication;
- length or complexity;
- photogenic spectacle;
- theoretical vocabulary;
- explicit self-criticism;
- clarity when clarification merely removes productive ambiguity;
- difficulty merely because it is difficult;
- feasibility achieved by weakening the artistic proposition.

These negative commitments matter because generic refinement has a predictable direction: add, clarify, technologize, activate, explain, professionalize. A dossier can become rhetorically more complete while its dominant operation becomes less necessary.

Ethical and public-context concerns are handled alongside this surrogate through review and eligibility reasoning rather than being introduced as an eleventh equally weighted aesthetic dimension. When a proposal involves testimony, surveillance, coercion, vulnerable groups, contested histories, labor, ecology, grief, public memory, or participation, these concerns can constrain what counts as an admissible or responsible realization. This separation avoids pretending either that ethical responsibility is irrelevant to artistic practice or that ethical virtue can simply be added numerically to artistic quality. Debates between aesthetic and dialogical accounts of participatory practice make precisely this kind of scalar synthesis difficult to defend [26, 36, 37].

2.4 Why Proposal Improvement Is Non-Monotonic

AIrtist is explicitly designed to improve proposals. The claim is not that art cannot be improved, nor that revision is inherently suspect. The difficulty is that open-ended artistic development lacks the kind of reliable local improvement signal available in tasks with a reference answer, a calibrated loss, or an objectively measurable defect. The problem is not the absence of an optimization target; it is the absence of a reliable local gradient toward that target. We do have an empirical objective: we want the resulting proposals to receive stronger independent holistic judgments. What we reject is a different assumption: that a correct criticism automatically identifies a revision that moves the proposal toward that objective. A critique may correctly identify a weakness and still motivate the wrong repair.

A proposal can become:

- more feasible but artistically less necessary;
- more immediately legible but more didactic;
- more interactive but less specific about spectatorship;
- more technologically responsive but more generic as an installation format;
- more historically differentiated but less coherent;
- visually stronger while weakening the material operation;
- more explicit while closing an ambiguity that previously sustained interpretation.

This is what we mean by non-monotonic improvement. The target remains better proposal quality. What fails is the assumption that every locally plausible revision supplies positive progress toward it.

Ambiguity as a special case Ambiguity makes the problem especially visible, but it is not the only source of non-monotonicity. Eco's Open Work offers one influential account of artistic structures that permit multiple realizations or interpretations without collapsing into arbitrariness [32]. Gaver, Beaver, and Benford similarly argue in design that ambiguity can function as a resource for interpretation rather than merely as missing information [38]. Interactive-art research warns that evaluation borrowed from conventional engineering can misread experiences whose value depends on reflection rather than task completion [39]. The implication for AIrtist is not: ambiguity is always good. An incoherent or underspecified proposal can hide behind ambiguity just as easily as a didactic proposal can over-explain itself.

The more precise point is: A productive ambiguity leaves something unresolved while still organizing what a spectator can perceive, infer, or experience. Such a property is difficult to improve through an instruction like make this clearer. Clarification may remove noise. Or it may remove exactly the unresolved relation that gave the proposal interpretive pressure. Participation produces an analogous trade-off. Bishop and Ranciere give different reasons to reject the assumption that converting spectators into visible actors automatically constitutes artistic progress [26, 25]. A revision that adds visitor input may increase activity while weakening a previously precise spectator position.

From critique to candidate intervention The architectural consequence is therefore specific: A diagnosis is evidence that a proposal may need intervention; it is not evidence that the first intervention proposed in response to that diagnosis is better. The same distinction gives the experiment a clear empirical target. We can ask not merely whether the system produces revised text, but whether:

- the final sampled possibility set is preferred to increasingly strong controls;
- and the interventions actually selected by the system are preferred to the exact parent states they displaced.

2.5 Maintaining a Sampled Possibility Set

The same uncertainty changes the appropriate output of ideation. Computational-creativity research describes creativity in terms of exploring and transforming spaces of possibilities rather than producing one isolated artifact [40, 41, 42]. Design-fixation research provides a complementary warning: early examples can constrain subsequent concepts even when alternatives are explicitly requested [43]. Large language models make the tension especially visible. Generating five fluent proposals does not guarantee five serious directions. A model can repeatedly vary the same metaphor, object type, interaction format, or rhetorical framing while changing vocabulary. For the intended use case, one polished answer is therefore an unnecessarily narrow target. AIrtist keeps five proposal lineages alive and returns all five surviving dossiers to the artist. We call this finite population the sampled possibility set. The phrase is deliberately modest. Five proposals are not the possibility space of contemporary art, do not exhaust the generative model, and are not claimed to be the optimal search width. They are the alternatives the system has actually constructed and can therefore expose to a user and evaluate.

This population view makes distinctions observable that winner-only evaluation collapses. A system may:

- occasionally produce one outstanding proposal while leaving the rest of the set weak;
- improve several candidates simultaneously;
- narrow the set while increasing its measured quality;
- generate a strong option but rank it poorly;
- produce several incompatible but serious directions for later human selection.

The quality of the available set and the ability of the system to surface one member of that set are therefore different empirical objects.

Sampled-set quality is not diversity Several candidates should not be mistaken for a diverse candidate set. A pool can be broad but weak. Another can be narrow but strong. Diversity itself can refer to semantic difference, artistic operation, material strategy, visual form, spectator position, political stance, art-historical distance, or some combination of these. Parallel lineages and heterogeneous challenger roles are architectural mechanisms intended to resist premature collapse. They are not empirical evidence that diversity actually increased. AIrtist therefore separates sampled-set quality from direct diversity. The main evaluation asks whether the measured proposal set becomes stronger. Latent-semantic breadth is evaluated separately and post hoc. This distinction becomes empirically important in our results: improved proposal quality does not imply greater semantic spread.

2.6 Scope of the Computationalization and the Empirical Target

AIrtist uses an explicit internal surrogate of proposal quality to structure generation, critique, preservation, intervention, and ranking. We do not equate that surrogate with artistic truth.

Textual dossiers as proxies for future works The first working assumption is representational: A sufficiently developed textual dossier can specify an installation hypothesis well enough for proposal development and comparative judgment. The dossier is expected to contain enough information to reason about the proposition, dominant artistic operation, visual and spatial encounter, material behavior, spectator relation, field position, characteristic failure modes, and production conditions. It is not expected to function as a fabrication specification. Engineering, site negotiation, conservation, artist decisions, prototype testing, material contingencies, installation practice, and audience behavior remain outside the textual representation or are represented only provisionally. Accordingly, our strongest empirical claims concern textual commission hypotheses, not realized artworks.

The internal model is a surrogate, not the outcome itself AIrtist uses the ten-dimensional surrogate to organize search, while the primary outcome is a separate held-out holistic preference judgment. The experiment therefore tests whether search guided by the internal surrogate transfers to a broader external measure rather than counting success under the surrogate as its own evidence.

External evaluation is intentionally more holistic The primary GPT-5.6 Sol outcome therefore does not ask the judge to reproduce AIrtist's ten-dimensional score vector, apply its dimension weights, or mechanically score every internal criterion. The external evaluator sees:

- the commission brief and its constraints;
- the common production context;
- Candidate A and Candidate B as complete anonymized proposal dossiers.

It is then asked to choose which proposal is, overall, the more compelling, distinctive, coherent, visually and spatially convincing, responsive to the brief, and plausibly realizable contemporary museum installation. The protocol explicitly states: Do not mechanically score dimensions or require all qualities to be equally present. It also instructs the judge not to reward length, fluency, polish, explicit theory, or self-criticism by themselves. Thus the external instrument is not norm-free. It still expresses a broad request for holistic artistic judgment.

What external Sol judgment does—and does not—stand for GPT-5.6 Sol remains a model-based judge rather than a proxy for professional curatorial consensus or realized-artwork value. Its role is to provide a holistic external measure distinct from AIrtist's ten-part internal surrogate.

We test: internal quality surrogate $\rightarrow$ structured search $\rightarrow$ external holistic model preference. We do not yet test: external model preference $\rightarrow$ artist or curator preference $\rightarrow$ quality of a realized installation. Those are distinct construct- and ecological-validity questions.

Computational objective The resulting computational problem can now be stated without either minimizing or overstating what the system attempts to do: Given an artist-authored brief, construct several sufficiently specified installation hypotheses; use an explicit bounded surrogate of proposal quality to diagnose and search for changes expected to improve them; require each proposal to articulate a dominant operation that can be preserved or challenged; allow strong states to survive when proposed interventions fail to earn adoption; maintain several alternatives rather than collapsing immediately to one answer; and test whether the resulting proposals receive stronger holistic preference under a separately instantiated external evaluator. This is not a claim to have solved artistic value computationally. It is a claim that a normative model of artistic proposal development can be made explicit enough to generate falsifiable computational hypotheses—and that its success or failure can be evaluated outside the ten-dimensional vocabulary from which the search itself was organized.

3 Related Work and Positioning

The term concept denotes different objects across the literatures adjacent to this work. In image generation, it can mean an object category or semantic attribute; in professional concept design, it often denotes a visual direction for a character, environment, or product. Here, concept development refers to the formation of the artistic proposition of a work: what the installation does, and how its material, spatial, spectator, historical, and institutional relations constitute that proposition. This distinction is important because recent AI systems already support sophisticated forms of ideation and creative generation, but usually operate on a different creative object.

We review three relevant lines of work. Contemporary-art practice shows generative systems moving upstream from artifact production toward artistic dialogue and the conception of possible works. Research on AI-assisted ideation establishes structured exploration and iterative development as computational problems in adjacent domains. Agentic systems further provide mature mechanisms for critique, branching, and selection. Together, these literatures motivate the task studied here while leaving the early development of a contemporary-artwork proposition largely unformalized as an evaluated computational problem.

3.1 AI in Contemporary-Art Practice

Contemporary-art practice has already begun to move generative AI upstream from the production of finished images into the formation of possible artworks. Agnieszka Kurant's Errorism (2021) provides a particularly direct precedent. GPT-2 and GPT-3 were trained on descriptions of Kurant's previous works and her essays and used to generate descriptions of conceptual works that she had never made but could potentially create; Kurant subsequently visualized this speculative material as holographic animations [3]. The significance of Errorism for the present work is not simply that an artist used a language model. The model operates at a moment when a future artwork is still being articulated: it extrapolates possible works from an existing artistic practice rather than merely rendering a specification supplied by the artist.

Other recent practices extend this relation between generative systems and artistic conception. In Alexander Reben's I'm Not A Painter (2022), a custom neural network and GPT-3 generate descriptions of nonexistent artworks; Reben curates these outputs, passes selected descriptions through text-to-image systems such as DALL-E 2, and selects images for physical realization [44]. Jonas Lund describes the works in In the Middle of Nowhere (2023) as having been conceptualized and produced through extended conversations with ChatGPT [4]. Such projects make language models part of a process of proposing, negotiating, and selecting possible works, rather than using them solely as image-making tools.

By 2024, related practices were also appearing as persistent, artist-specific systems. Cai Guo-Qiang's cAI is a custom multimodal model that learns from Cai's artworks, archives, and interests and is positioned by his studio as a partner for dialogue and collaboration [5]. cAI participated in the conception and choreography of the Getty-commissioned WE ARE (2024) [6]. Anicka Yi's Emptiness project similarly uses an algorithm trained on years of her studio's output as a "digital twin" of her practice [45]. Holly Herndon and Mat Dryhurst's The Call (2024–25) makes another aspect of this shift explicit by treating the construction of vocal datasets, governance frameworks, and choral AI models as part of artmaking itself [46].

These practices make the motivating use case for AIrtist substantially less speculative. Established artists are already allowing generative systems to propose possible works, participate in artistic direction, and develop persistent computational counterparts to their practices. At the same time, these are artist-specific artistic procedures. Their systems are embedded in a particular oeuvre, studio process, exhibition, or authored experiment; they are not intended to accept an unfamiliar artistic problem and systematically develop several complete responses to it.

Recent empirical studies of artists working with generative AI reinforce this distinction. A preregistered experiment with professional visual artists shows that artistic expertise remains consequential when working with text-to-image AI [47], while qualitative work with contemporary artists characterizes human–AI co-creativity as a sustained, relational process shaped by artistic intuition, control, and adaptation to a system's affordances [48]. Such findings motivate treating AIrtist as one bounded computational component of a larger artistic process: the artist supplies the problem and remains responsible for subsequent artistic development, while the present system investigates what computational support can contribute between those stages.

A small number of computational-art projects have also begun to place LLM agents inside contemporary-art discourse. Artism, for example, combines a simulated social network of virtual artist agents with a system that recombines fragments of contemporary-art discourse into synthetic movements, visualizations, and criticism. Its purpose is primarily the simulation and critique of patterned artistic production rather than development of an artist-authored project, and the reported evidence is a preliminary case study rather than a controlled test of artwork concept development [49].

The trajectory across these practices is therefore clear: AI participation in artistic conception already exists. What remains missing is a general computational formulation of this activity that can be specified, inspected, and evaluated across artistic problems rather than within one artist's practice.

3.2 AI-Assisted Ideation and Design-Space Exploration

Early-stage ideation is considerably more mature as a computational problem in adjacent creative and professional domains. AIdeation, for example, supports professional concept designers in film, television, and games through brainstorming, reference exploration, and recombination. Its development is grounded in a formative study with 12 professional designers and evaluated through a controlled study with 16 professionals and a subsequent deployment in four studios [7]. The work provides strong evidence that generative systems can usefully intervene before a visual design has stabilized rather than merely execute a completed specification.

CoExploreDS addresses a related property of ill-defined design problems: problem formulation and solution development can co-evolve. The system represents emerging problems and candidate solutions explicitly and uses multiple reasoning modes to propose further directions [50]. Its relevance to AIrtist lies in establishing structured exploration of an evolving creative state as a serious design problem in its own right.

Research ideation offers an autonomous analogue. ResearchAgent decomposes scientific ideation into literature-grounded problem identification, method and experiment generation, review, and iterative revision [8]. The AI co-scientist similarly maintains and develops multiple hypotheses through generation, reflection, ranking, evolution, and meta-review [9]. Si et al., comparing LLM-generated research ideas with ideas written and blindly reviewed by more than one hundred NLP researchers, further show that idea generation and idea evaluation need not improve together: their agent produces ideas judged more novel on average while exposing failures in self-evaluation and a lack of diversity [51].

These systems establish that AI-assisted ideation, structured exploration, multi-candidate development, and iterative criticism are not new in themselves. The distinction in AIrtist concerns the state being developed. Visual concept design ultimately produces production assets;

product design converges toward functional solutions; research ideation produces hypotheses and experimental plans. In contemporary-art concept development, changing a material relation, spatial arrangement, spectator position, or institutional procedure can change the identity of the proposed work itself. This motivates the domain representation introduced in Section 2 rather than treating the task as generic brainstorming with an art-themed prompt.

A complementary body of work also cautions that generative assistance does not necessarily broaden exploration. Controlled studies report stronger fixation under exposure to generated visual examples [10], greater semantic similarity across LLM-assisted ideators [52], and improvements to individual outputs accompanied by reductions in collective diversity [11]. Recent work on multi-agent ideation likewise shows that denser interaction and larger groups can produce diminishing diversity returns and premature convergence rather than broader exploration [53]. These findings make the existence of multiple candidates a poor proxy for the quality or breadth of the resulting possibility field.

The Cultural Alien Sampler provides an art-generation-specific example of explicitly measuring such exploration. Its computational object, however, differs sharply from that studied here. CAS deliberately reduces its search space to combinations from a finite vocabulary of 3,500 visual concepts—examples include sequences such as ukiyo-e + machinery + hunter—which are subsequently converted into prompts and rendered by an image generator. Its relevance is methodological rather than task-level: the paper shows that different generative policies exhibit measurably different repetition, exploration-radius, and saturation behavior over repeated generations [54].

This literature motivates two aspects of our evaluation. First, the retained candidate population should be treated as an object of analysis rather than assuming that producing several outputs constitutes meaningful exploration. Second, candidate quality and candidate breadth are distinct properties: a system can improve one without necessarily improving the other.

3.3 Agentic Revision and Selection

Iterative critique, role specialization, and selection are established components of current agentic systems. ResearchAgent uses collaborative reviewing agents before revising scientific ideas [8], while the AI co-scientist combines reflection and evolutionary development with tournament-based ranking among hypotheses [9]. More general inference-time methods such as Self-Refine and Tree of Thoughts replace a single completion with iterative feedback or exploration over multiple candidate reasoning states [55, 56].

Role-specialized refinement also appears in creative-generation systems. CREA, for example, coordinates Creative Director, Prompt Architect, Art Critic, and Refinement Strategist agents for creative image editing and generation. Its task remains explicitly image-level—the system transforms or generates objects such as couches, cars, and other visual categories—and its critic evaluates rendered images against generic creativity dimensions before further regeneration [57]. CREA therefore demonstrates the maturity of multi-agent creative refinement as a mechanism, rather than providing a precedent for the development of an artwork proposition.

AIrtist builds on the general premise that additional inference can be allocated to critique and alternative generation, but changes the status of revision. In canonical iterative-refinement methods such as Self-Refine, feedback is used to produce a revised successor [55]; ResearchAgent likewise feeds reviewer feedback into iterative revision [8]. For the task studied here, identifying a weakness does not imply that the available repair is preferable to the existing proposal. Review is therefore first converted into an asymmetric local state: what gives the incumbent its strongest identity, what most needs intervention, which familiar operating logic should be left, and what forms of repair should not count as progress. Focused and Structural challengers are generated independently from the same parent, while the unchanged incumbent remains eligible.

The resulting empirical question differs from ordinary endpoint refinement: when the system actually elects to transform an artistic proposal, is the adopted intervention preferable to the exact state it was intended to improve? Exact-parent comparison makes that question directly observable.

Selection creates a second separable problem. Research-ideation results already suggest that generative ability can exceed reliable self-evaluation [51]. AIrtist therefore does not infer ranking quality from proposal quality. Its terminal selectors act on exactly the same generated five-candidate set and are evaluated against an external utility unavailable during selection. This isolates failure to recognize a strong candidate from failure to generate one in the first place.

3.4 Positioning AIrtist

The preceding literature leaves a specific task gap. Contemporary-art practice already demonstrates language models and other generative systems participating upstream of final realization, including the proposal of hypothetical artworks and ongoing artist-specific computational collaborators. Adjacent computational research already provides sophisticated methods for ideation, structured exploration, critique, revision, and ranking. Yet across the systems and documented artistic practices reviewed here, we did not identify an evaluated general-purpose system whose primary task is to take a previously unseen, under-specified artist-authored contemporary-art installation brief and develop several complete hypotheses for what the artwork itself could become.

To our knowledge, AIrtist is the first evaluated computational system designed specifically for this task. Its outputs are not image concepts or thematic suggestions, but complete project hypotheses in which an artistic operation is articulated together with its material, spatial, spectator, field, and institutional realization.

AIrtist further proposes a particular computational treatment of concept development. Complete project hypotheses persist as competing states; critique specifies both preservation and transformation constraints; proposed changes compete with the incumbent rather than automatically replacing it; and evaluation distinguishes the quality of the resulting proposal set, the consequences of adopted interventions, and the effectiveness of terminal selection.

Contemporary-art installation ideation is therefore the substantive domain of the work, while also exposing a broader issue for open-ended creative agents. Failing to construct strong alternatives, damaging an existing state through revision, and failing to recognize strong material that has already been generated are different failure modes. The experiments in this paper are designed to make those failures separately observable. The next section gives the agentic computational formulation used to study this task.

4 AIrtist: An Agentic Architecture for Installation Concept Development

AIrtist takes a single artist-authored brief and returns five developed installation proposals in ranked order. In the evaluated configuration, the artist supplies no intermediate feedback: proposal construction, review, branching, competition, and ranking all occur internally after the initial brief is submitted. The five returned dossiers should therefore be understood as ranked starting points for subsequent artistic development, not as finished artworks.

AIrtist represents concept development through persistent proposal lineages rather than as a chain of increasingly polished rewrites. Computationally, those lineages induce a local-search process over complete proposal states: each maintains an incumbent proposal, constructs diagnostic evidence about it, converts that evidence into a constrained local search hypothesis, generates two differently instructed challengers from the unchanged parent, and allows the incumbent to survive if neither challenger earns replacement. A terminal ranker operates only after the five lineages have been developed.

The architecture therefore separates three kinds of state that generic "critique and rewrite" systems often collapse:

$$\begin{aligned}\text{proposal state} &\xrightarrow{\text{specialized review}} \text{diagnostic evidence}\\ &\xrightarrow{\text{opportunity memo}} \text{local search hypothesis}\\ &\xrightarrow{\text{focused / structural}} \text{candidate interventions}.\end{aligned}$$

A review is not itself a revision instruction. An opportunity memo is not a revised proposal. And a generated challenger does not automatically replace its parent.

Figure 1 summarizes the end-to-end architecture. Proposal construction and field-context construction begin as separate streams. They first meet at specialized review and opportunity synthesis. Competitive refinement then proceeds independently across five lineages before a separate terminal ranking stage orders the five surviving dossiers. Table 5 summarizes the interfaces. "Output" denotes the principal object passed forward rather than every piece of recorded provenance or diagnostic metadata.

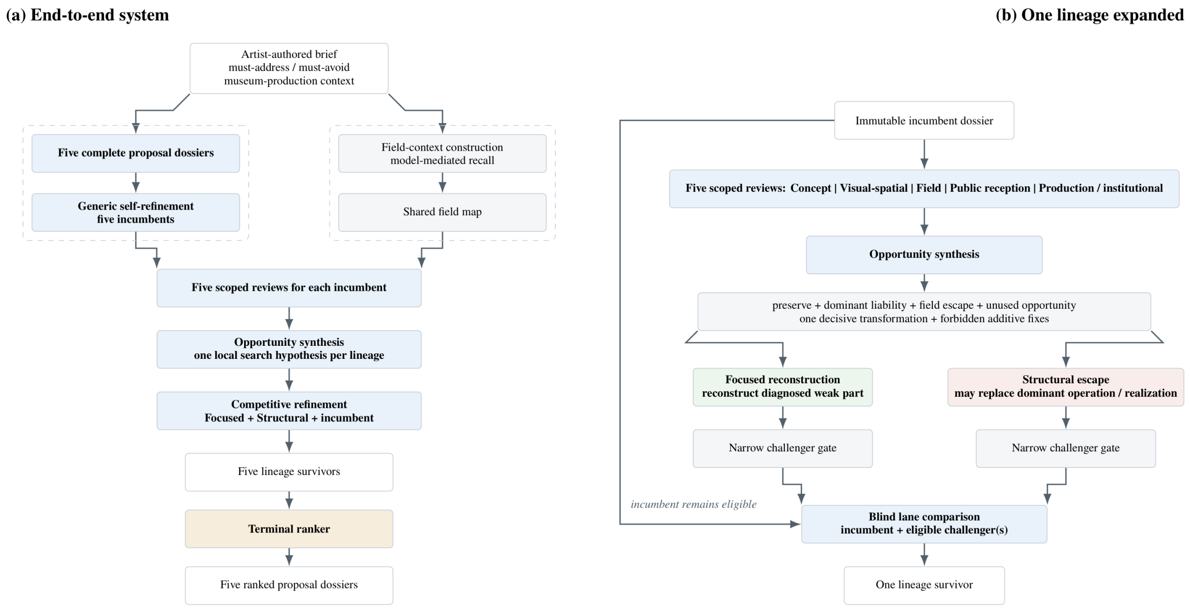

Figure 1: AIrtist architecture. (a) End-to-end flow from one artist-authored brief to five ranked proposal dossiers. Proposal-state construction and field-context construction are initially separate; the shared field map enters specialized review and opportunity synthesis. (b) One lineage expanded. Reviews produce diagnostic evidence; opportunity synthesis compresses it into one local search hypothesis; Focused and Structural challengers are independently generated from the unchanged parent. Only challengers are gated. The incumbent bypasses the gate, remains eligible, and competes with all eligible challengers in the blind lane comparison. The terminal ranker is a separate stage over the five lineage survivors.

Table 5: Stage interfaces in AIrtist.

Stage	Principal input	Principal output
Initial proposal construction	brief, explicit constraints, museum/production context, internal proposal-quality surrogate	five complete installation dossiers
Generic self-refinement	one initial dossier, brief and constraints, shared quality guidance	five developed incumbents
Field-context construction	brief and explicit constraints	shared field map
Specialized review	incumbent dossier, brief, field map, reviewer-specific scope	five scoped diagnostic packets
Opportunity synthesis	immutable incumbent, review packets, field evidence	compact local search hypothesis
Challenger generation	immutable incumbent, brief, opportunity memo	focused reconstruction + structural escape
Challenger eligibility	each new challenger, brief and hard constraints	gate eligibility decision
Lane comparison	incumbent + eligible challenger(s)	one surviving dossier per lineage
Terminal ranking	five survivors, brief and constraints	total ordering of all five dossiers

4.1 Input and Persistent Proposal State

4.1.1 Artist brief and commissioning context

The input to AIrtist is a partially specified artistic problem rather than a request for a finished artwork. The intended user is an artist who has a rough direction, knows some conditions that matter, and may know several solutions that should explicitly be avoided, but does not yet know what form the installation should take. AIrtist represents this input through three elements:

- a free-text project brief;
- explicit must-address conditions;
- explicit must-avoid conditions.

The free-text component states the artistic or situational problem. Must-address conditions

Table 6: Exact fields of the installation dossier.

Dossier field	What it represents
`title`	A working identifier for the proposal. It may contribute to framing but is not itself treated as an artistic operation or quality dimension.
`artistic_thesis`	What the work proposes, tests, places under pressure, or asks us to encounter. In Section 2 terminology, the dossier's concise articulation of artistic stakes.
`artistic_operation`	The dominant action or dependency through which the installation enacts the thesis rather than merely stating it.
`concept_description`	An integrative description connecting thesis, operation, visual/spatial/material realization, spectator relation, and institutional situation.
`visual_logic`	What is concretely visible or perceptible: dominant image or material presence, scale, silhouette, surface, light, color, texture, and viewing conditions.
`spatial_logic`	How route, position, access, distance, sequence, threshold, orientation, concealment, visibility, waiting, and bodily movement organize the installation.
`material_logic`	What materials, media, processes, absences, or realization rules physically, temporally, historically, socially, or institutionally do; which properties are constitutive and which may vary.
`spectator_encounter`	The spectator's position, attention, movement, action or non-action, access, duration, visibility, knowledge, and consequence within the installation.
`difference_from_field`	The proposal's own hypothesis about how it differs from nearby artistic formats or precedents; not independent evidence of art-historical novelty.
`risks_and_failure_modes`	The proposal's own account of how the work could become generic, decorative, gimmicky, incoherent, didactic, technically fragile, or otherwise lose artistic pressure.
`institutional_feasibility`	Site, staffing, maintenance, accessibility, safety, permissions, liability, equipment, construction, and production dependencies required for the commission hypothesis to operate.

identify pressures that should remain active rather than being replaced by an easier neighboring theme. Must-avoid conditions remove unwanted solution families without prescribing what should replace them. A prohibition on cameras, for example, removes one realization mechanism; it does not specify what a non-camera installation should be.

Unless the brief specifies another context, AIrtist assumes a temporary or medium-duration contemporary-art museum commission. The proposal may occupy a gallery, foyer, threshold, circulation area, stairwell, entrance zone, or service-adjacent museum space, but it is assumed to remain within a museum-controlled footprint. An unspecified brief is not silently reinterpreted as a city-street, municipal-infrastructure, or other civic project outside ordinary museum authority. The default production context assumes an approximate USD 150,000 project envelope covering project-specific fabrication, materials and equipment, specialist engineering and temporary technical labor, installation and deinstallation, safety measures, maintenance requirements, project-specific staffing, and contingency. As Section 2 emphasized, this is a reasoning context for a textual commission hypothesis, not evidence that engineering or fabrication has been validated.

Together these elements define the search problem: what artistic pressure must remain active, which solutions are inadmissible, and within what institutional world must a proposal function? They do not prescribe a medium, object type, interaction model, or artistic style.

4.1.2 The installation dossier as persistent authored state

The object manipulated by AIrtist is a structured installation dossier. It is the persistent authored state of a proposal lineage. Reviews, field evidence, and opportunity memos are attached to or derived from that state; they do not silently rewrite it. Challengers are generated from an immutable copy of the incumbent, and any selected challenger receives a traceable parent relation. This separation is what later allows the experiment to compare a selected intervention against the exact textual parent it displaced.

The implementation uses the following authored fields. To avoid ambiguity, we use the exact field names consistently throughout the paper.

Several distinctions in this representation matter downstream. difference_from_field is candidate-authored positioning. It should not be confused with the independently constructed field context described in Section 4.2. risks_and_failure_modes is likewise the proposal's own anticipation of failure. It should not be confused with the later independent reviewer failures. And institutional_feasibility represents the dossier's production hypothesis. It is later pressure-tested by a separate Production/Institutional reviewer. Section 2 defines the conceptual role of artistic operation and the relationship among stakes, operation, and realization. The dossier makes those distinctions explicit enough for later agents to diagnose them separately. A compelling thesis can coexist with arbitrary spatial logic; a visually specific proposal can use interchangeable material; a materially interesting object can establish no precise spectator relation. Keeping these fields separate allows the system to ask what should remain invariant and what can be reconstructed.

4.2 Initializing Search: Five Proposal Lineages and a Shared Field Context

AIrtist prepares two inputs for specialized development: a five-proposal candidate population and a shared hypothesis about the artistic field surrounding the brief. Both derive from the brief, but the streams are initially separate. Field context does not determine which five initial proposals are generated; it enters their development later, through review and opportunity synthesis.

4.2.1 Five proposal lineages

The first stream produces five complete installation dossiers in one batched generation. Each should constitute a distinct proposal rather than a continuation or paraphrase of another. Each dossier then undergoes one generic self-refinement pass. The pass may revise any authored dossier field while preserving the underlying proposal and strengthening brief fidelity, operation-realization coherence, spatial and material specificity, spectator encounter, and plausible museum realization. Initial proposal construction and this generic self-refinement already operate under installation-specific quality guidance. They do not yet receive the synthesized field map or specialized review packets introduced below.

All five self-refined proposals are retained. They become five incumbents, one per persistent lineage. AIrtist therefore does not select a single concept and recursively improve it. It begins specialized search with five complete hypotheses whose histories remain independent.

4.2.2 Field-context construction

In parallel, AIrtist constructs contextual evidence intended to help later stages recognize familiar artistic attractors and possible routes away from them. This stage is model-mediated recall, not provenance-bearing retrieval. It does not search a verified external bibliography. Recalled artworks, artists, critical writing, and neighboring cultural or material references serve as associative context for search and must not be interpreted as factual evidence that a proposal is globally novel in contemporary art.

The recall process considers four complementary kinds of material:

1. concrete artworks;
2. artists and collectives;
3. art writing, including criticism, catalogues, interviews, and exhibition discourse;
4. relevant material outside art, including cultural, historical, spatial, legal, technical, or social contexts.

Relevance is deliberately broader than topical similarity. A reference may matter because of material behavior, image logic, object presence, atmosphere, spectator position, temporal structure, institutional relation, bodily scale, sound, stillness, fiction, or another transferable pressure.

The field synthesizer converts this material into five explicit output categories:

- nearby_lines – nearby artistic approaches or operating logics;
- overused_formats – familiar uses that may become generic if repeated mechanically;
- productive_differences – dimensions along which the proposal could depart meaningfully;
- taboo_zones – locally dangerous attractors or especially obvious forms of repetition;
- field_insights – more specific observations grounded in the recalled material.

These are working search hypotheses. overused_formats does not prohibit a medium, taboo_zones does not create new must-avoid constraints, and field_insights does not certify art-historical facts. The same synthesized field context is shared across all five lineages.

4.3 Specialized Multi-Agent Review: Diagnosis as Evidence

An installation dossier can fail in several independent ways. A compelling proposition may remain spatially arbitrary. A precise spatial situation may rely on an interchangeable material. An otherwise coherent work may reproduce a familiar artistic format. Its visible public signal may collapse into a gadget or awareness display. Or its operation may depend on hidden staffing, inaccessible circulation, or an unsafe physical mechanism. AIrtist therefore reviews each incumbent through five deliberately scoped diagnostic agents. The split is not intended to simulate a curatorial committee voting on artistic value. It decomposes AIrtist's ten-dimensional internal quality surrogate from Section 2 into non-overlapping diagnostic ownership.

Table 7: Review agents and ownership of the internal surrogate.

Reviewer	Owned internal dimensions	Primary diagnostic question
Concept	raw-prompt fidelity; core-tension enactment; artistic-operation necessity; material necessity	Does the proposal enact the brief through one necessary dominant operation, or could its form be replaced without substantial artistic loss?
Visual-spatial	spatial encounter; visual encounter; spectator specificity	What does a spectator concretely see, occupy, cross, wait for, avoid, or experience, and how do position and duration change that encounter?
Field	field distinctiveness	Under the model's recalled field context, does the proposal transform a nearby artistic logic or merely reproduce an available format?
Public reception	public-reception resilience	What plausible public reading could flatten the proposal into a gimmick, spectacle, awareness device, wellness environment, decorative object, or generic engagement format?
Production / institutional	production/institutional viability	Can the stated artistic relation plausibly operate with respect to fabrication, safety, accessibility, maintenance, staffing, permissions, liability, and the shared production envelope?

4.3.1 Review packet interface

Each reviewer receives:

- the incumbent dossier;
- the original brief and explicit constraints;
- relevant shared field evidence;
- reviewer-specific scope instructions.

It returns diagnostic evidence rather than a rewritten proposal.

A review packet contains:

- strengths;
- failures;
- revision_demands;
- diagnostic severity;
- diagnostic repairability;
- diagnostic confidence;
- scope_handoff_notes when a concern primarily belongs to another reviewer.

The Production/Institutional reviewer additionally returns a structured production-envelope assessment when enough information is present.

The numerical diagnostics are not votes. They are not averaged across reviewers to decide whether a proposal survives. In the frozen review contract, failures and revision demands are explicitly the primary review output; severity, repairability, and confidence summarize the diagnosis within that reviewer's restricted scope. An empty failure list is valid. The stage can therefore be written as:

$$D_i \xrightarrow{\text{five scoped reviewers}} R_i = \{r_i^{\text{concept}}, r_i^{\text{visual}}, r_i^{\text{field}}, r_i^{\text{reception}}, r_i^{\text{production}}\}.$$

The result is evidence about a proposal, not five competing rewrites.

Running example: Delayed Projection (Brief 035) The brief asks for a work about public shame while excluding cameras, mirrors, humiliation performance, and moralizing text. The incumbent Delayed Projection nevertheless uses infrared cameras to capture visitor movement and replay delayed silhouettes onto a semi-transparent scrim. Concept reviewer. It identifies the delayed silhouette as a genuine strength: temporal displacement turns the public into an absent presence. But it also identifies a direct must-avoid violation: the proposal depends on infrared cameras and surveillance imagery. Its revision demand is structural and specific: replace live capture with a pre-programmed generative silhouette loop. Visual-spatial reviewer. It treats the delayed silhouettes and the scrim bisecting the room as strong spatial-perceptual elements and records no in-scope failure. Field reviewer. It likewise regards temporal displacement as a meaningful modification of surveillance imagery and records no field failure, while handing the camera prohibition back to the appropriate scope. Public-reception reviewer. It predicts two plausible reductive readings: the delayed silhouettes may be received as a technological gimmick or as an Instagrammable photographic backdrop. Production / institutional reviewer. It flags visitor recording as a privacy and consent liability. Importantly, its immediate repair proposal is largely administrative: add signage, opt-out provisions, and a data-retention policy. This disagreement is exactly why review is separated from revision. If AIrtist concatenated all reviewer demands and told a model to "fix everything," it could keep the camera system and add signage, consent procedures, explanatory text, and other compensations. The dossier would become longer without resolving the artistic problem created by live surveillance. Instead, review evidence is passed to a separate synthesis stage.

4.4 Opportunity Synthesis: From Diagnosis to a Local Search Hypothesis

The opportunity-synthesis agent receives the immutable incumbent dossier, its five review packets, and the shared field context. It does not rank candidates and does not write a revised proposal. Its function is to transform heterogeneous diagnosis into one tractable search hypothesis. We use *opportunity state* for this conceptual local-search state and *opportunity memo* for the concrete recorded schema through which that state is passed to downstream challenger generation. This stage is deliberately lossy. Specialized reviews may expose several valid weaknesses, possible repairs, production cautions, and field similarities. Attempting to satisfy all of them simultaneously would encourage additive revision. Opportunity synthesis instead decides which problem should organize the next search move and what must survive while that problem is addressed.

The actual memo schema contains six substantive fields plus evidence references.

Table 8: Opportunity memo schema. Formal field names are paired with plain-language roles.

Exact field	Plain-language role
`incumbent_strength_to_preserve`	Preserve: the strongest property that should not be accidentally destroyed by the next intervention.
`dominant_artistic_liability`	Liability: the weakness that should organize the local search step.
`field_similarity_to_escape`	Field escape: a nearby operating format or artistic attractor that the next move should avoid reproducing mechanically.
`unused_material_or_spatial_opportunity`	Opportunity: an underused material or spatial degree of freedom available to the proposal.
`one_decisive_transformation`	Search anchor: one concrete transformation worth testing rather than a list of independent repairs.
`forbidden_additive_fixes`	Do not substitute: additions that should not be accepted as a substitute for addressing the central liability.
`evidence_ids`	Traceability back to the diagnostic evidence used in synthesis.

The phrase one decisive transformation can sound as if synthesis has already decided how the proposal must be rewritten. That is not its computational role. The memo proposes a shared intervention hypothesis that both challenger agents will test under different search permissions. It is not an accepted revision. The stage therefore answers: What should the next candidate try to change, and what should it protect while doing so? It does not answer: Which new proposal should replace the incumbent?

4.4.1 Brief 035 opportunity state

The actual memo for Delayed Projection is especially informative:

The memo resolves the reviewer conflict by preserving temporal delay while treating recording itself—not its administration—as the search problem. This illustrates why review packets are evidence rather than revision instructions.

4.5 Competitive Refinement

For each lineage i, AIrtist generates two challengers independently from the same immutable incumbent x_i and the same opportunity memo m_i:

$$L_i = \{x_i, c_i^{\text{focus}}, c_i^{\text{escape}}\}.$$

Table 9: Opportunity state for the Brief 035 running example.

Memo field	Recorded state
`incumbent_strength_to_preserve`	temporal delay that produces a ghostly, absent audience and makes withholding of presence felt
`dominant_artistic_liability`	live infrared cameras violate the surveillance prohibition and create privacy liability
`field_similarity_to_escape`	familiar recorded-surveillance installation logic
`unused_material_or_spatial_opportunity`	a floor fissure or material rupture that can make exclusion bodily without recording visitors
`one_decisive_transformation`	replace live capture with a pre-programmed generative silhouette loop while preserving temporal delay
`forbidden_additive_fixes`	mirrors, live camera feeds, moralizing explanation, literal waiting line

The focused challenger is not generated from the structural challenger, and the structural challenger is not generated from the focused one. This independence is important. A structural branch cannot merely inherit incidental additions introduced by the focused branch; both are separate responses to the same parent and the same diagnosis.

4.5.1 Focused reconstruction versus structural escape

The distinction between the two challenger roles is procedural rather than categorical. The frozen prompts define them as follows. A focused reconstruction:

- preserves the artistic problem and the memo-protected strongest element;
- is organized around the diagnosed weak component;
- may deeply reconstruct the weak artistic operation, material logic, spatial logic, visual logic, or spectator relation;
- attempts to remove the dominant liability rather than compensate for it through explanatory or administrative additions.

A structural escape:

- preserves the brief, artistic problem, hard constraints, and memo-protected strength;
- is explicitly permitted to replace the dominant artistic operation or realization when remaining close to the incumbent would preserve the diagnosed field similarity;
- must be materially different from the incumbent.

This means that a focused reconstruction is not forbidden from changing artistic operation. The distinction is instead one of search intent and permission: Search permission, not edit size. Focused: reconstruct the diagnosed weak part while remaining organized around the incumbent's protected identity. Structural: if local reconstruction would preserve the same problematic artistic logic, permission extends to replacing how the proposal works as a whole. The difference resembles refactoring versus substitution, but even that analogy should not be taken too literally. The outputs are generated proposals, not deterministic transformations with formally guaranteed edit radii.

4.5.2 Same parent, same memo: what "reconstruct" and "replace" look like

Brief 044 provides a useful controlled example because both challenger agents receive the same parent and same opportunity memo. The parent Weighted Light Path encodes unequal waiting through: pressure pad $\rightarrow$ fixed delay $\rightarrow$ LED ribbon illuminates. The opportunity memo says to preserve fixed, touch-triggered hierarchical delays while identifying the purely visual LED flash / light-up-floor trope as the liability and field similarity.

The two challenger modes respond differently. Focused reconstruction – Delayed Glow Wall. The focused branch keeps the same operative grammar: pressure $\rightarrow$ fixed delay $\rightarrow$ light becomes visible. But it reconstructs the weak visual-material realization. Instead of an LED ribbon simply turning on, a delayed UV pulse excites a phosphorescent strip that glows briefly. The liability is addressed inside the existing operation: the unequal delay still culminates in light, but the visible event and material process are rebuilt.

Structural escape – Delayed Reveal Wall. The structural branch retains the artistic problem and the fixed hierarchy of delay, but abandons light as the dominant visible event: pressure $\rightarrow$ fixed delay $\rightarrow$ mechanical panel swings open. A concealed color panel physically reveals itself after the programmed latency.

Table 10: Same parent and memo, different transformation permissions in Brief 044.

	Focused	Structural
Trigger	pressure	pressure
Core temporal problem	unequal fixed delay preserved	unequal fixed delay preserved
Dominant event	reconstructed light event: phosphorescent afterglow	replaced event: mechanical reveal
Search move	rebuild weak realization from inside	leave the lighting realization altogether

Here the branch uses its broader permission to replace the dominant realization and perceptual event rather than reconstruct the lighting system. The example also shows why we should not define the roles simply as "small change" versus "large change." The focused branch can substantially rebuild a component. The structural branch can sometimes remain recognizably close to the parent while replacing the component that organizes the work.

4.5.3 The distinction is a permission, not a guarantee

The roles are not guaranteed to produce different semantic distances. In Brief 030, for example, the focused and structural agents independently converge on almost the same solution family: both replace the clock-mediated pull of Chronicle of Pull with unequal reclaimed-brick masses directly loading a shared veil.

This convergence is not inconsistent with the architecture. Structural escape is permission to replace more, not a requirement to produce maximal novelty. When the opportunity memo and brief strongly constrain a plausible repair, two independently generated roles may converge.

4.5.4 Challenger eligibility

Generating a challenger does not make it a legitimate competitor. Each new challenger first passes a narrow eligibility gate. The incumbent is not re-gated and remains available throughout the lane. Deterministic checks remove exact duplication of the parent or sibling challenger. The model-based gate is reserved for narrow invalidity conditions such as:

- malformed output;
- negligible material difference from the incumbent;
- explicit must-address or must-avoid failure;
- loss of brief fidelity;
- another concrete hard-constraint violation;
- severe safety failure.

Ordinary artistic weakness, unfamiliarity, production complexity, cost within the broader development envelope, or incomplete readiness are not by themselves hard-gate conditions. This distinction connects directly to Section 2: the gate defines eligibility; it is not another component of the ten-dimensional internal quality surrogate. A proposal may be eligible and still artistically weak. Such a candidate should lose through comparison rather than disappear through a disguised aesthetic veto.

4.5.5 Blind lane competition

The incumbent and all gate-eligible challengers then enter lane competition. Candidate origin is hidden from the internal selector. The comparison sees authored dossier content, the brief and explicit constraints, and AIrtist's ten-dimensional internal surrogate of proposal quality. It does not receive reviewer packets, opportunity memo, candidate role, lineage history, or prior selection outcome. This prevents the lane selector from rewarding a challenger merely because earlier agents invested effort in producing it.

The update has three cases.

1. No challenger passes the gate. The incumbent survives automatically.
2. Exactly one challenger passes.

Incumbent and challenger receive independent pointwise assessment under the ten-dimensional internal surrogate. The higher-scoring proposal survives, with deterministic resolution of an exact tie.

1. Both challengers pass. All three candidates receive pointwise assessment and are additionally ranked under three cyclic presentation orders so that each candidate occupies each list position once. These rankings are aggregated through Borda evidence and first-place counts, followed by pointwise score and a deterministic final tie-break.

No incumbent bonus is used. Thus preservation is possible without being privileged. One proposal survives each lineage. Repeating the procedure independently across five incumbents produces the five-proposal set passed to terminal ranking. The distinction between this internal decision and the paper's external evaluation is important. The lane selector acts under AIrtist's internal ten-dimensional surrogate. The later GPT-5.6 Sol outcome used in the experiment is the held-out holistic comparison introduced in Section 2.6 and is unavailable to the search system.

4.6 Terminal Ranking

Competitive refinement answers: Which state should survive within each proposal lineage? It does not answer: In what order should the artist inspect the five surviving proposals? AIrtist therefore uses a separate terminal ranker.

This distinction matters because the final interface exposes all five dossiers but does not expose them with equal salience. Rank 1 acts as the system's strongest recommendation and can frame the artist's first impression of the candidate set. The terminal ranker receives the five survivor dossiers, the original brief, and explicit constraints. It is blind to search history. It does not receive reviewer packets, field context, opportunity memos, incumbent/challenger roles, lane-selection decisions, or earlier internal scores.

4.6.1 Pointwise screening

All five proposals are first assessed independently under the same ten-dimensional internal surrogate used for proposal-quality comparison elsewhere in AIrtist. The assessment yields ten quality scores, a decisive strength, a decisive liability, confidence, and readiness risks.

This stage also supports a narrow terminal hard veto, distinct from ordinary low quality. The veto is restricted to conditions such as severe explicit-constraint violation, physically impossible central mechanism, unsafe failure required for the work to function, or a safety defect that cannot be removed without replacing the central operation. Ordinary production difficulty, maintenance burden, staffing, familiarity, or a safely remediable accessibility problem do not automatically trigger a veto. A vetoed proposal is not deleted from the five-proposal output. It remains in the final ordering below admissible candidates and cannot be promoted into the finalist tournament. Terminal eligibility therefore affects rankability at the top of the list, not candidate-set size.

4.6.2 Top-of-pool tournament

Up to the three highest-scoring non-veto proposals become finalists. If fewer than three admissible candidates exist, the tournament shrinks rather than promoting a vetoed candidate simply to fill a fixed shortlist. Every finalist pair is judged twice with exact presentation reversal. For finalists a and b, one comparison presents a as Candidate A and b as Candidate B; the second presents the same dossiers in the opposite order. The pairwise comparator receives the brief, explicit constraints, and the two authored dossiers. It does not receive the candidates' pointwise scores or search history. The two orientations are mapped back to candidate identity before aggregation.

Across finalist pairs:

1. logical wins and losses determine Copeland score;
2. accumulated orientation-level pairwise points provide the next ordering signal;
3. pointwise quality score breaks remaining ties;
4. a deterministic content-based rule resolves an exact residual tie.

Candidates outside the finalist set retain their pointwise order below the tournament finalists; vetoed proposals remain below admissible proposals. The final product is therefore a total ordering of all five surviving dossiers, not a shortlist and not a single surviving concept.

4.6.3 Why use a tournament?

The motivation is not that pairwise comparison is assumed to be intrinsically superior. Pointwise screening asks a broad question independently of the other candidates: How strong is this proposal under the internal surrogate? The tournament concentrates additional inference on a narrower question: When the strongest-looking candidates are compared directly, does their relative order change? This architecture was designed to concentrate expensive comparative judgment near the top of the fixed pool.

Whether that additional comparison actually improves the external quality of the first-ranked proposal is not assumed by the architecture. RQ3 evaluates exactly that question on identical generated candidate sets. The 50-brief experiment therefore treats the tournament as a tested ranking policy, not as an architectural component whose superiority is taken for granted.

4.6.4 System output

The final output presented to the artist is a ranked set of five complete proposal dossiers. The ranking is best interpreted as an allocation of attention among starting points, not as a claim that the first dossier constitutes the artwork or that lower-ranked proposals have no value. The architecture consequently preserves two objects at the end of search:

1. a fixed, traceable population of five developed installation hypotheses;
2. a separate internal policy for ordering that population.

This separation lets the evaluation distinguish the developed proposal population from the policy that orders it, without treating the top-ranked dossier as the only system output.

5 Experimental Setup

The experiment separates three empirical objects: RQ1 evaluates the retained five-proposal set, RQ2 evaluates adopted interventions against their exact parents, and RQ3 evaluates terminal ranking while holding all proposal texts fixed. A separate selected-output campaign provides a secondary end-to-end check.

Table 11: Evaluation decomposition. The rows isolate candidate-set quality, intervention quality, and ranking quality rather than treating the final top-ranked proposal as a single undifferentiated outcome.

Analysis	Empirical object	Held fixed	What changes	Main outcome
RQ1: sampled-set quality	Five retained proposals	Brief, generation model, external judging protocol	Search condition	Complete-pool preference / centered effect
RQ2: competitive intervention	Individual proposal lineage	Exact incumbent parent	Whether and how the lineage is transformed	Selected replacement vs. exact parent
RQ3: terminal selection	The same five A5 proposals	Every proposal text	Terminal ranking policy only	External Utility@1
Secondary end-to-end check	One surfaced proposal	Brief and holistic judging protocol	System whose selected output is shown	Holistic pairwise preference

5.1 Benchmark

Held-out commission briefs The benchmark contains 75 commission-style installation briefs. Briefs 001–025 were available during system development and are excluded from every reported evaluation. The empirical study uses the remaining 50 held-out briefs, IDs 026–075. The released benchmark is the frozen benchmark_75_v1 dataset; the reported held-out slice contains ten briefs in each of five coverage categories.

Each brief represents a partially specified artistic problem rather than a request to reproduce an existing artwork or optimize toward a reference answer. As described in Section 4, it contains three kinds of information: a free-text project premise, a must-address field that identifies pressures the proposal should preserve, and a must-avoid field that removes unwanted solution families. Unless the brief itself specifies another setting, proposals inherit the common contemporary-art commission and production context defined in Section 4.

Table 12: Held-out benchmark coverage. Categories are sampling devices, not proposed artistic genres. Examples reproduce the frozen benchmark semantics in shortened form.

Coverage category	What it stresses	Representative held-out example
Thematic / conceptual	Translating an abstract or social pressure into an operative installation rather than an illustration	Brief 026: public apologies continuously revised before they can be heard; revision must become a shared material burden rather than a written statement.
Site-specific	Making an existing spatial condition constitutive while preserving its practical use	Brief 037: unequal public ascent through an underused municipal stairwell, using handrails, landings, and daily wear while preserving every safety requirement.
Material-constrained	Producing an operation from a restricted material vocabulary	Brief 038: disputed land boundaries using only chalk, reclaimed brick, and rainwater-safe sealant; the work must remain readable after partial weathering.
Institutional / production-constrained	Working within access, staffing, maintenance, or public-use conditions	Brief 044: unequal waiting in a museum foyer during school visits; the work must tolerate touching and require no attendant beyond normal front-of-house staff.
Anti-cliche / difficult	Blocking highly available first-association solutions	Brief 035: public shame while excluding mirrors, cameras, humiliation performance, and moralizing text, with a precise passerby relation to what is withheld.

The must-address and must-avoid fields are important to the experiment. They prevent brief fidelity from being reduced to thematic keyword overlap. For example, a proposal can remain nominally "about" public shame while violating a prohibition on cameras, or address unequal access while replacing the specified spatial condition with a generic symbolic object. Conversely, a must-avoid condition restricts a solution family without prescribing what should replace it. The benchmark therefore tests development under both positive and negative constraints.

The categories are coverage devices, not proposed artistic genres and not a taxonomy of contemporary installation practice. The benchmark was curated within this project rather than sampled from the naturally occurring distribution of museum commissions. It therefore supports controlled comparison across a deliberately varied set of constraints, not an estimate of performance on contemporary-art proposals in general. Similarly, the population size of five proposals is an experimental and system-design choice rather than an asserted optimum. Five is large enough to maintain several simultaneous project hypotheses while still permitting exhaustive cross-pool evaluation. The object evaluated below is therefore a finite sampled possibility set, in the sense defined in Section 2, rather than an approximation to the full space of artistically viable directions.

5.2 Experimental Conditions

All core generation and internal decision conditions use openai/gpt-oss-120b through OpenRouter, routed exclusively to Groq with provider fallback disabled. The frozen run uses temperature 0.8 across model-mediated stages. An experiment-level random seed of 42 controls deterministic ordering and related reproducibility mechanisms where the implementation uses a seed; it should not be interpreted as making stochastic model sampling deterministic. Candidate generation and all internal system decisions are frozen before external judging begins. The conditions are not four unrelated generations. They form a deliberately nested control structure.

Raw five-proposal pool (B5-raw) B5-raw is the five-dossier population produced by the initial batched generation, before any self-refinement or specialized search.

Generic self-refinement (B5-self) Each of the five raw dossiers then undergoes the same single generic self-refinement pass used to initialize AIrtist. The resulting five proposals form B5-self. This condition controls for an important simple alternative: perhaps most of the benefit attributed to agentic search can be obtained by asking the same model to inspect and improve each initial proposal once. B5-self is also more than a baseline: these five exact proposal texts become the persistent incumbents from which both AIrtist and the matched iterative control begin. Consequently, the later A5–M5 contrast is not confounded by one system receiving a luckier initial sample. Both start from the same five self-refined parent dossiers with the same lineage identities.

Matched generic iteration (M5) M5 is the central mechanism control. Starting independently from each of the same five B5-self parents, it maintains five proposal lineages, generates two challengers per lineage, leaves the incumbent eligible, applies the same narrow challenger gate, and uses the same blind lane-comparison procedure as AIrtist. What it removes is the domain-structured organization of local search. M5 does not receive AIrtist's specialized multi-role review, field-conditioned opportunity state, explicit preserve/liability/escape representation, or differentiated focused-reconstruction and structural-escape roles. Its two challengers instead perform generic proposal development. M5 therefore tests a stronger explanation than

"AIrtist makes more calls." Both systems perform multi-candidate local iteration from identical parents and select among old and new states. A residual A5–M5 difference is consistent with a contribution from the package of domain-structured search mechanisms rather than from additional sampling alone. It is important, however, not to overstate this control. M5 is substantially matched, not compute-identical. Exact prompt lengths, contextual inputs, call counts, and token expenditure are not equal. The A5–M5 contrast therefore does not estimate improvement per token or per model call, and it cannot identify which reviewer, opportunity-state field, preservation constraint, or challenger role is individually necessary or sufficient.

AIrtist retained set (A5) AIrtist begins from the same B5-self parents and applies the architecture described in Section 4: field-context construction, five specialized reviews per lineage, opportunity synthesis, independent focused and structural challengers, challenger gating, and competitive lane selection. One proposal state survives each lineage, producing a five-proposal set denoted A5.

Table 13: Nested control structure for the candidate-set experiment. A check mark means that the condition includes the corresponding property.

Experimental property	B5-raw	B5-self	M5	A5
Five complete initial dossiers	✓	✓	✓	✓
One generic self-refinement pass per dossier	–	✓	✓	✓
Same five B5-self parent texts	–	–	✓	✓
Five persistent lineages	–	–	✓	✓
Two challengers per lineage	–	–	✓	✓
Incumbent remains eligible	–	–	✓	✓
Same challenger gate	–	–	✓	✓
Same blind lane-comparison procedure	–	–	✓	✓
Specialized multi-role review	–	–	–	✓
Field-conditioned opportunity synthesis	–	–	–	✓
Explicit preserve / liability / escape state	–	–	–	✓
Focused and structural challenger roles	–	–	–	✓

5.3 Blind External Evaluation

Evaluation instrument The primary external evaluator is GPT-5.6 Sol, used in fresh ChatGPT Temporary Chats. It is separate from the gpt-oss-120b model used for generation and internal system decisions. The evaluator is instructed not to browse the web, use prior project context, or infer which system produced either candidate. For each comparison, the evaluator receives the original brief, its explicit must-address and must-avoid conditions, the common museum-commission context, and two complete anonymized proposal dossiers labelled Candidate A and Candidate B. The commission context assumes a temporary or medium-duration contemporary-art museum commission with an approximate production envelope of USD 150,000. The envelope covers project-specific fabrication, materials, equipment, engineering, temporary technical labor, handling, installation/deinstallation, permits, safety measures, consumables, maintenance materials, project-specific labor, and reasonable contingency; it excludes artist fee, international transportation, taxes, and ordinary permanent museum guards.

The external judging protocol is deliberately holistic. It asks which candidate is, overall, the more compelling, distinctive, coherent, visually and spatially convincing, responsive to the brief, and plausibly realizable contemporary museum installation. A tie is explicitly allowed. The evaluator is instructed not to reward length, fluency, polish, explicit theory, or self-criticism by themselves, and not to mechanically score dimensions. This distinction matters because the internal system and terminal rankers use the ten-dimensional bounded working model introduced in Section 2, whereas the external Sol protocol is not a copy of that scoring sheet. The two instruments overlap in the artistic concerns they attend to, but external pairwise utility is derived from a separate holistic judgment task rather than from AIrtist's own internal component scores. The external evaluator does not receive system identity, source condition, proposal lineage, incumbent or challenger status, internal scores, internal rank, reviewer output, opportunity state, previous judging results, or the result of the opposite presentation orientation. External candidate utilities computed from these judgments are never returned to the generator, lane selector, or terminal rankers.

Complete cross-pool matrices For every brief, each of the five A5 candidates is compared with every candidate in each control pool. Thus A5 versus one control contributes 5 × 5 = 25 logical candidate comparisons per brief. Across 50 briefs and three control pools, 50 × 3 × 25 = 3,750 logical comparisons are evaluated. This exhaustive design is central to RQ1. Evaluating only the top output of each system would entangle candidate generation with internal ranking and would make a strong system appear weak if its best candidate happened to be ranked poorly. The complete matrix instead makes the measured quality of every candidate observable independently of the system's own selector.

Exact A/B reversal Every logical comparison is presented twice with exact candidate-order reversal. If one presentation shows proposal x as Candidate A and y as Candidate B, the paired presentation shows the same texts as y versus x. The primary evaluation therefore contains 3,750 × 2 = 7,500 physical judgments. Each response contains an A/B/tie verdict, confidence on a 1–5 scale, and a short comparison-specific justification. Confidence is recorded for diagnostics but never used to weight primary utility. The swapped design does not make the two judgments statistically independent; its purpose is to make presentation-order instability observable. For execution, each campaign/pass bundle is evaluated in a fresh Temporary Chat, with multiple comparisons processed within that context. Pass A and Pass B are evaluated in separate fresh chats. The 7,500 judgments should therefore be understood as dense repeated measurement over 50 experimental tasks, not as 7,500 independent samples.

5.4 From Pairwise Judgments to Candidate-Set Utility

For one physical presentation, a verdict is mapped back to the identity of candidate x:

$$v(x,y) = \begin{cases} 1, & x \text{ preferred,} \\ 0.5, & \text{tie,} \\ 0, & y \text{ preferred.} \end{cases}$$

Let $v_1(x, y)$ and $v_2(x, y)$ denote the two exact swapped presentations after both are mapped back to the same candidate identity. Their average defines the canonical logical-pair score

$$s(x,y) = \frac{v_1(x,y) + v_2(x,y)}{2},$$

so that

$$s(x, y) \in \{0, 0.25, 0.5, 0.75, 1\}.$$

A value of 1 means both orientations prefer x; 0 means both prefer y. A value of 0.5 can arise either from two ties or from compensating orientation-sensitive outcomes. The intermediate 0.25 and 0.75 values occur when a tie in one orientation accompanies a directional preference in the other.

For an AIrtist candidate a_i evaluated against a five-candidate control pool B,

$$U_B(a_i) = \frac{1}{5} \sum_{b_j \in B} s(a_i, b_j).$$

This quantity is the candidate's external utility against that opponent pool. It is external in the operational sense that AIrtist's search and ranking procedures never observe it.

The corresponding complete-pool preference for one brief is

$$P(A5 > B) = \frac{1}{25} \sum_{i=1}^{5} \sum_{j=1}^{5} s(a_i, b_j),$$

which is equivalently the mean external utility of the five A5 candidates against that control pool.

For interpretability we also report a centered representation,

$$\Delta U = 2P(A5 > B) - 1.$$

Here, $\Delta U = 0$ denotes parity between the two five-proposal populations, $+1$ would mean that every swapped comparison favors A5 in both orientations, and -1 the reverse. Preference and centered effect are not two independent metrics. They are two scales for the same complete 5 × 5 comparison matrix.

5.5 Measures for the Three Research Questions

The metric hierarchy is deliberately explicit. We distinguish the primary outcome of each experiment from secondary diagnostics that characterize the same underlying candidate set or ranking. A secondary statistic becoming significant does not retroactively turn it into an additional primary hypothesis.

RQ1: sampled-set quality The primary RQ1 outcome is the brief-level mean complete-pool effect ΔU defined above. It is computed separately for A5 vs. B5raw, A5 vs. B5self, A5 vs. M5. The first two comparisons measure improvement relative to simpler stages of the same generation process; the third is the central shared-parent mechanism comparison.

Mean pool quality alone, however, does not tell us how improvement is distributed among the five proposals. A system could increase its average because one exceptional candidate becomes much stronger while the remaining proposals stagnate or deteriorate. We therefore report four additional oracle distributional diagnostics. For each pool, candidate utilities are computed against the opposing five-candidate pool. We then identify:

- Best-candidate utility: the utility of the externally strongest candidate in the five-proposal set.
- Top-three mean utility: the average external utility of the three strongest candidates.
- Worst-candidate utility: the utility of the externally weakest candidate.
- Top-heaviness: the gap between the externally best candidate's utility and the mean utility of its own pool.

Differences between A5 and each control on these quantities describe whether the pool shift is concentrated at the top, extends through several candidates, or reaches the lower tail.

The word oracle is important here. The identities of the externally best, top-three, and worst candidates are determined after judging from external utility. They are not AIrtist's internal ranks and do not measure whether the system knew which candidate was best. That question belongs to RQ3. These distributional quantities are secondary diagnostics of RQ1. They are not additional members of the primary three-contrast hypothesis family.

RQ2: competitive intervention RQ2 separates what the refinement policy does from whether its adopted changes improve over what they replace. The first quantity is descriptive. Across all five lineages for every brief, we record whether the final proposal state is content-identical to the B5-self incumbent, a selected focused reconstruction, or a selected structural escape. An unchanged final state should not be interpreted as a semantic "Keep" decision in every case: it can also occur because no challenger remains eligible after gating. We therefore describe this statistic conservatively as the proportion of final lane states that remain identical to the incumbent. The stronger RQ2 test concerns elected replacements. Because A5 and B5-self share exact parent identities and the A5–B5-self evaluation is exhaustive, every selected challenger already has a blind external comparison against the specific incumbent proposal from which it was generated.

For a selected replacement c and its exact parent p, the parent-relative score is $s(c, p)$, and its centered form is

$$\delta(c, p) = 2s(c, p) - 1.$$

A positive value favors the selected challenger; a negative value favors the displaced parent. Because some briefs contain several selected replacements while others contain none, the primary inferential quantity is brief-balanced and conditional on intervention. We first average the centered exact-parent effects of all selected replacements within a brief. Briefs in which no lineage is replaced do not contribute to this conditional replacement-quality estimand; their behavior is represented instead by the separate retention statistic. Statistical inference is then performed across the resulting brief-level replacement effects.

We additionally report raw replacement preference and role-specific outcomes for focused and structural replacements. These role-specific figures are descriptive. Focused and structural challengers are generated and selected under different local circumstances, so a higher observed preference for one role does not by itself establish a causal advantage of that operator. RQ2 therefore supports a deliberately bounded inference: it can test whether interventions actually adopted by the frozen policy tend to beat their exact parents. It does not compare AIrtist with a counterfactual policy forced to replace every incumbent, because that counterfactual candidate set was not generated.

RQ3: terminal selection on an identical candidate set RQ3 removes generation as a source of variation entirely. For every brief, the Base selector and the Hybrid tournament selector rank exactly the same five frozen A5 proposal texts. Neither selector observes the external Sol judgments used for evaluation. The Base selector is the frozen content-only terminal ranker used in the ablation. It receives the brief, constraints, and five authored dossiers, scores each proposal on the same ten equal-weight internal quality dimensions, applies the narrow terminal hard-veto tier, and orders admissible candidates by their mean pointwise score. The bounded top-two comparator is disabled in this condition. The Hybrid selector is the terminal policy described in Section 4. It preserves the same content-only information boundary but adds an independent pointwise stage followed by pairwise comparison among up to the three highest-scoring non-veto finalists. Finalist pairs are judged under exact A/B reversal and aggregated into a tournament ordering before the complete five-item order is returned. Thus RQ3 changes the ranking policy, not the candidate texts available to it.

To score either selector, each A5 candidate is assigned an aggregate external utility over all 15 baseline opponents:

$$U(a_i) = \frac{1}{15} \sum_{b \in B5_{\text{raw}} \cup B5_{\text{self}} \cup M5} s(a_i, b).$$

Thus the external relevance signal used for RQ3 incorporates all three complete cross-pool matrices while remaining unavailable to the selectors. Let $\pi = (\pi_1, \ldots, \pi_5)$ denote a selector's ordering of the five A5 candidates. The primary RQ3 outcome is

$$\text{Utility@1}(\pi) = U(a_{\pi_1}),$$

the external utility of the proposal the selector places first. This metric directly captures the attention-allocation problem motivating terminal ranking: given the exact same proposal set, does one selector surface an externally stronger first option?

We additionally report several secondary ranking diagnostics. Regret@1 measures the external utility sacrificed relative to an oracle top choice:

$$\text{Regret@1}(\pi) = \max_i U(a_i) - U(a_{\pi_1}).$$

For a fixed candidate pool, Regret@1 is algebraically coupled to Utility@1; it is useful because it expresses the top-ranking error as lost utility, but it is not an independent confirmation of a Utility@1 effect. Exact top-1 accuracy asks whether the proposal ranked first attains the maximum external utility in its pool. If several candidates share the maximum, selecting any of them counts as correct. Hit@2 and Hit@3 ask whether at least one externally best candidate occurs within the first two or first three positions. NDCG@5 treats external utility as graded relevance and evaluates the quality of the complete five-candidate ordering, giving greater weight to errors near the top. Utility@1 is the primary selector endpoint. Regret, top-k coverage, and NDCG characterize different practical aspects of the ranking but are not treated as a collection from which whichever metric happens to reach significance can substitute for the primary outcome.

Table 14: Metric hierarchy. Post-hoc semantic analyses are intentionally excluded from the primary metric set and are introduced where used in Results.

Analysis	Primary inferential quantity	Secondary / descriptive quantities
RQ1	Mean complete-pool ΔU for three planned control contrasts	Best, top-three, worst, top-heaviness; brief W/T/L
RQ2	Brief-balanced selected-replacement effect vs. exact parent	Incumbent-identical / focused / structural frequencies; pair preference; role-specific outcomes
RQ3	Paired difference in Utility@1 on identical A5 pools	Regret@1, Top-1, Hit@2, Hit@3, NDCG@5
Selected-output check	Holistic selected-pair preference	Directional consistency with complete-pool results
Measurement robustness	No new primary endpoint	Swap stability, position, consensus-only, verbosity, and leave-out sensitivity

Analyses introduced after inspection of the frozen experiment—most notably latent-semantic pool breadth, the relationship between quality gain and breadth, and parent-relative semantic distance between focused and structural challengers—are not part of this primary metric set. They are introduced explicitly as post-hoc diagnostics at their first use in the Results and Mechanism Analysis sections.

5.6 Statistical Inference

The brief is the statistical unit throughout the primary analysis. This point is important because exhaustive comparison produces many judgments from each task. For RQ1, one brief contributes 25 logical candidate comparisons for each control, and each logical comparison is observed in two presentation orientations. These observations provide a denser estimate of the candidate sets associated with that brief; they do not convert one artistic problem into dozens of independent benchmark tasks. Accordingly, pair-level scores are first aggregated to the relevant brief-level statistic before uncertainty is estimated.

Confidence intervals use 10,000 brief-level bootstrap resamples. Resampling is performed over briefs rather than over individual candidate pairs. For paired hypotheses, we use a paired randomization test at the brief level: under the null of no systematic directional effect, the sign of each brief-level paired difference is randomly reversed. The resulting reference distribution is used for two-sided significance testing.

The three planned RQ1 complete-pool comparisons $A5 - B5_{\text{raw}}$, $A5 - B5_{\text{self}}$, $A5 - M5$ form one hypothesis family and are corrected using Holm's procedure. No larger omnibus correction is retroactively constructed across all quantities in the paper. RQ2 exact-parent improvement and RQ3 Utility@1 are distinct experiments with their own primary estimands. Distributional pool diagnostics, secondary ranking metrics, and robustness analyses retain their explicitly secondary or diagnostic status rather than being used as alternative routes to a positive primary claim.

For interpretability, we also report brief-level win/tie/loss counts. These are based on the sign of the corresponding brief-level effect and make heterogeneity visible even when the aggregate mean is positive. Judge-reported confidence never changes the primary score, the bootstrap weight, or the statistical unit.

5.7 Secondary Selected-Output Evaluation

Complete-pool evaluation deliberately removes the system's internal ranker from RQ1. It therefore answers a question about the quality of the available five-proposal set, not the more familiar end-to-end question of what happens if a user attends only to the proposal surfaced first. We examine that question in a separate blind selected-output campaign. For each held-out brief, we take the proposal ranked first by AIrtist's frozen Hybrid terminal selector and compare it separately with the top-ranked proposal from each of the three control conditions under their frozen ranking artifacts: A5top vs. B5self,top, A5top vs. M5top, A5top vs. B5raw,top. This yields three logical selected-output comparisons per brief, or 150 logical comparisons across 50 briefs. Each is again evaluated under exact A/B reversal, yielding 300 physical judgments.

The selected-output campaign is run separately from the complete-pool evaluation but uses the same blind holistic judging criterion and commission context. The external judge sees the brief, constraints, and two complete dossiers, but not system identities, source pools, candidate IDs, internal ranks, or generation histories.

This experiment is deliberately secondary. For one system contrast, complete-pool evaluation observes 25 cross-pool relationships per brief; the selected-output check observes only one. It consequently discards most of the information available about the sampled candidate set and reintroduces dependence on each system's internal ranking. Its value is instead interpretive. If the complete-pool advantage and the selected-output comparison point in the same direction, the former has not obviously been created by improvements that disappear as soon as the system is collapsed back to the single-output presentation a user is most likely to notice first.

5.8 Measurement-Robustness Analyses

The primary evaluation relies on a language-model judge, so the experiment separately examines several identifiable ways in which that measurement could mislead.

First, the exact-swapped design makes presentation-order stability observable. We report agreement between the two canonical A/B orientations as well as the aggregate preference for whichever proposal happened to be physically displayed as Candidate A. The latter should be near 0.5 in the absence of a gross global position preference.

Second, for the central A5–M5 comparison we perform a consensus-only sensitivity analysis, restricting the matrix to logical pairs for which the two exact orientations yield the same canonical verdict. This is not a cleaner replacement estimator: agreement filtering changes the population of comparisons and may preferentially retain easier cases. Its purpose is narrower—testing whether the central effect depends on the comparisons whose verdict visibly reverses under presentation order.

Third, we examine relative verbosity. Structured agentic pipelines may produce longer dossiers, and model judges can reward elaboration independently of substance. We therefore measure candidate length, test its association with pairwise preference, and examine the A5–M5 direction in comparisons where A5 is no longer than its opponent and where it is strictly shorter. These subsets are robustness diagnostics rather than additional confirmatory endpoints.

Fourth, we test dependence on benchmark composition through leave-one-brief-out and leave-one-category-out re-estimation of the central matched-control effect. These analyses ask whether the overall direction is explained by one unusually favorable task or one complete coverage category; they are not used to infer that AIrtist performs differently across artistic categories.

6 Results

6.1 RQ1: Domain-Structured Development Improves the Sampled Possibility Set

Table 15 reports the three planned complete-pool contrasts. For every brief and control condition, all five A5 proposals are compared with all five control proposals. Mean complete-pool preference therefore reflects the entire 5 × 5 relation between two sampled sets rather than either system's selected top output.

All three planned contrasts favor A5. The largest effect is against raw generation. A single generic self-refinement pass narrows the difference but does not remove it. Most importantly, A5 also exceeds M5, the substantially matched control that begins from the same five B5-self parent texts, maintains the same five lineages and two challengers per lineage, leaves the incumbent eligible, and shares the challenger-gating and lane-comparison procedures.

The A5–M5 comparison is therefore the central RQ1 result. Identical parents rule out a more favorable starting sample as the explanation. Matching much of the iterative scaffold also makes a simple "more continuations from the same parents" account substantially less plausible. The principal designed differences are AIrtist's domain-structured evidence, opportunity-state representation, preservation constraints, and differentiated focused and structural search roles. The residual +0.0836 effect consequently supports a package-level contribution from domain-structured local search relative to substantially matched generic iteration. It does not isolate the causal contribution of any individual component, nor does it estimate improvement per token or model call.

The aggregate result is heterogeneous rather than universal. Against M5, A5 has a positive brief-level effect on 31 of 50 briefs, ties on 8, and a negative effect on 11. Figure 2 exposes this distribution rather than reducing it to its mean. The leave-one-brief analysis reported below provides the stronger robustness statement: no single brief determines the sign of the matched-control estimate.

Table 15: Primary complete-pool results over 50 held-out briefs. All three planned contrasts remain significant after Holm correction.

Comparison	A5 pref.	ΔU	95% CI	Brief W/T/L	Holm p
A5 vs B5-raw	0.6068	+0.2136	[+0.1412,+0.2848]	42/0/8	$< .0001$
A5 vs B5-self	0.5502	+0.1004	[+0.0488,+0.1512]	32/11/7	.0008
A5 vs matched M5	0.5418	+0.0836	[+0.0340,+0.1304]	31/8/11	.0015

6.1.1 The gain is not confined to an oracle-best candidate

A higher five-proposal mean can still arise in different ways. Structured search might produce one exceptional proposal while leaving the rest of the pool unchanged or weaker, or it might shift a larger portion of the sampled set. Section 5 therefore defined the externally best candidate, top-three mean, externally worst candidate, and top-heaviness as secondary oracle diagnostics.

The matched M5 comparison is particularly informative. The oracle-best difference is positive but imprecisely estimated (+0.0560, 95% CI [-0.0020, +0.1150]). In contrast, the top-three mean differs by +0.0767 (unadjusted 95% CI [+0.0173, +0.1363]), and the externally worst proposal differs by +0.0710 ([+0.0250, +0.1180]). Top-heaviness does not increase (-0.0276, [-0.0670, +0.0114]).

These diagnostics are secondary and their intervals are not members of the Holm-corrected primary family. Nevertheless, their pattern is informative. The matched-control advantage is not confined to the oracle maximum: it is visible in the pool mean, the upper three candidates,

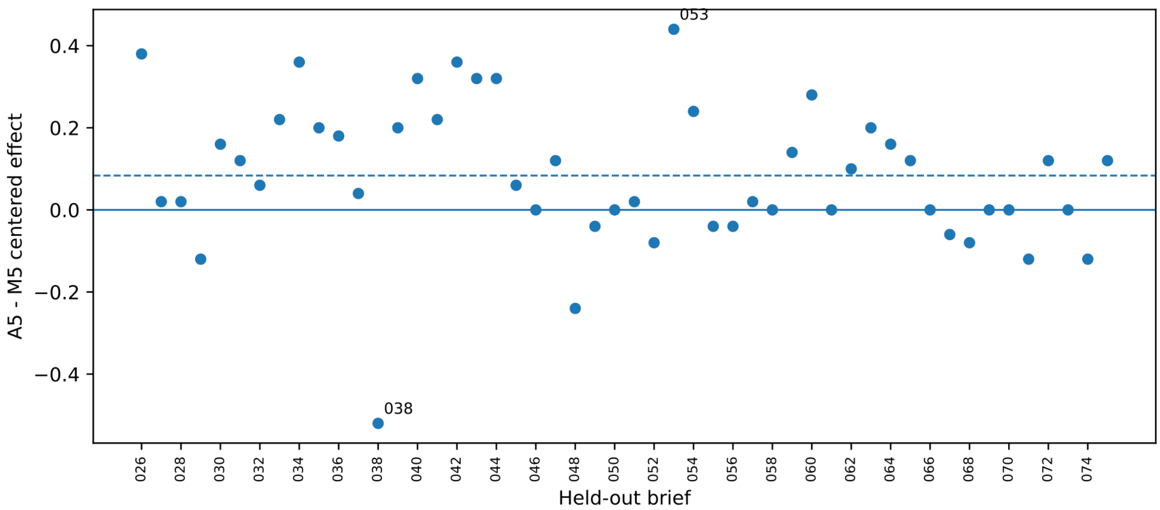

Figure 2: Matched-control effect across all 50 held-out briefs. Each point is one brief-level centered A5-M5 complete-pool effect; the dashed horizontal line is the overall mean. The aggregate effect is a distributional shift, not dominance on every commission problem. Briefs 038 and 053 are annotated because they return later as qualitative mechanism cases.

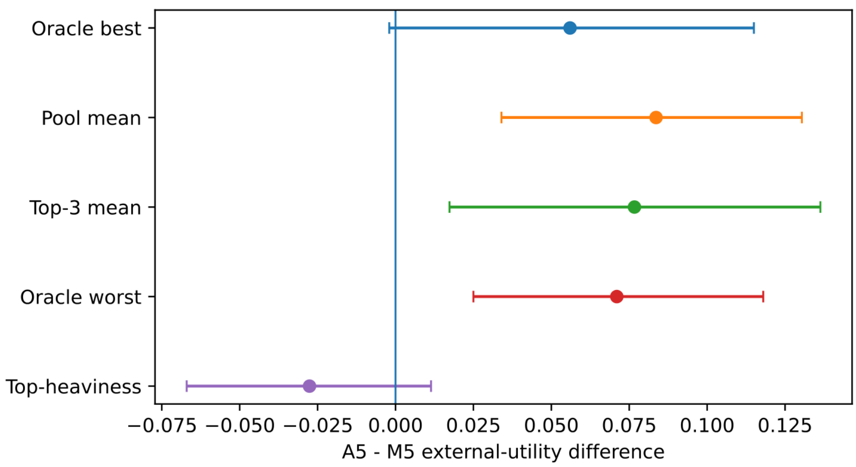

Figure 3: Where the A5-M5 difference occurs within the sampled possibility set. Points show brief-balanced differences with 95% brief-bootstrap intervals. Pool mean is the planned primary endpoint; the remaining quantities are secondary oracle diagnostics with unadjusted intervals.

and the lower tail, while there is no detected increase in top-heaviness. The pattern is therefore inconsistent with a purely top-heavy account in which the aggregate gain is carried only by an isolated lucky maximum.

Table 16: Distributional diagnostics for RQ1. These are secondary oracle descriptors of the externally evaluated pools, not selector ranks.

Control	Best Δ	Mean Δ	Top-3 Δ	Worst Δ	Top-heaviness Δ
B5-self	+0.0660	+0.1004	+0.1190	+0.0530	-0.0344
M5	+0.0560	+0.0836	+0.0767	+0.0710	-0.0276
B5-raw	+0.2060	+0.2136	+0.2123	+0.1750	-0.0076

6.1.2 Post-hoc: higher measured quality does not imply greater latent-semantic breadth

Section 2 deliberately separated sampled-set quality from breadth. Maintaining five lineages does not establish that they occupy a wider proposal space, and higher external preference does not logically imply greater diversity. We therefore examine this distinction post hoc.

Using the primary semantic representation—artistic operation, spatial logic, material logic,

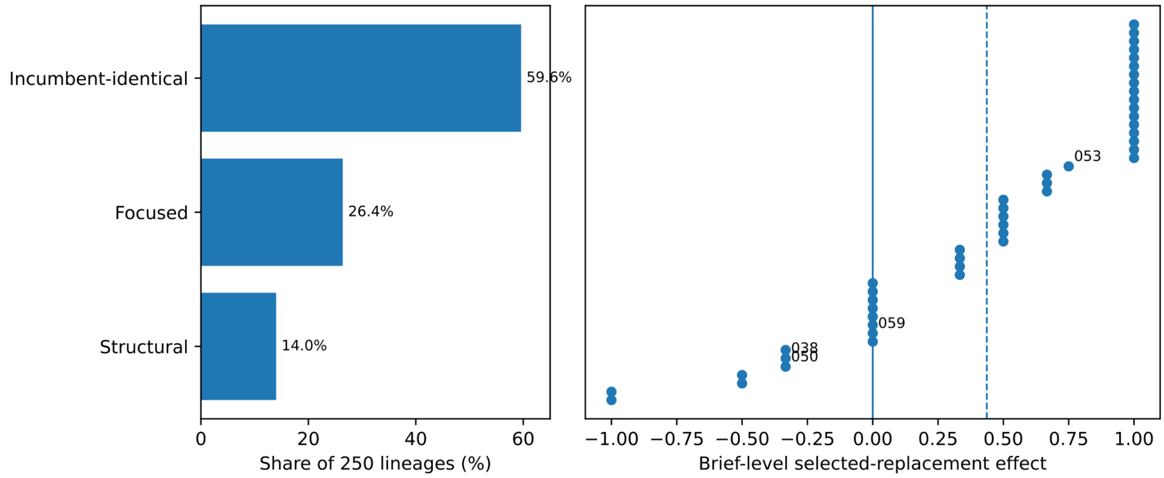

Figure 4: Competitive refinement separates whether a lineage changes from whether the adopted change helps. Left: final states over 250 lineages. Right: brief-level exact-parent effects among briefs with at least one selected replacement, sorted by effect; the dashed line is the brief-balanced mean. Labels mark cases used later in the mechanism analysis.

and spectator encounter—A5 pools have lower mean within-pool latent-semantic distance than M5 pools. The paired A5–M5 breadth difference is -0.0554 (95% CI [-0.0744, -0.0385]); A5 is more concentrated on 43 of 50 briefs. Adding concept description to the representation preserves the qualitative direction.

This finding should not be generalized to "diversity" without qualification. It concerns one explicitly defined latent-semantic representation. Under that representation, however, higher measured proposal quality does not imply greater breadth. At the brief level, Pearson correlation between quality gain and breadth difference is $r = -0.229$ ($p = .110$), while the exploratory rank association is Spearman $\rho = -0.331$ ($p = .019$). Twenty-nine briefs combine a positive A5–M5 quality effect with lower A5 breadth, whereas only two combine positive quality effect with greater breadth. We make no causal claim from this post-hoc association: it does not show that semantic concentration causes quality improvement or that wider search would be undesirable.

6.2 RQ2: Selected Interventions Tend to Improve on Their Exact Parents

Across 250 lineages, 149 final states remain content-identical to their incumbents (59.6%), 66 end in focused reconstructions (26.4%), and 35 in structural escapes (14.0%). Thus 101 lineages end in a new proposal state. The 59.6% figure is a property of the complete refinement pipeline rather than a count of explicit semantic "Keep" decisions: a final state can remain identical because challengers fail eligibility or because the incumbent survives subsequent competition.

Because A5 and B5-self retain exact lineage identity, every adopted challenger has an existing blind comparison with the specific parent text it displaced. Across 101 selected replacements, external preference for the challenger is 0.693. After averaging multiple replacement effects within each brief, the brief-balanced centered effect is +0.437 (95% CI [+0.266, +0.594], paired randomization $p < .0001$); brief W/T/L is 31/8/7.

Among 73 selected-replacement comparisons whose exact A/B orientations return the same verdict, challengers beat their parents 56 times and lose 17 times. The remaining 28 replacement comparisons are orientation-sensitive. The RQ2 result is therefore stronger than the descriptive retention frequency: conditional on the frozen policy adopting a replacement, selected challengers are externally preferred on average to the exact states they replace. The experiment does not contain a mandatory-replacement counterfactual and therefore does not establish that incumbent retention itself causes the gain.

6.2.1 Focused and structural roles produce different transformation magnitudes

The selected focused and structural replacements have parent-relative external preferences of 0.659 and 0.757, respectively. These values are descriptive because the two roles are generated and selected under different local conditions.

A more direct post-hoc question is whether the two prompts produce different magnitudes of movement from the same exact parents. Under the primary semantic representation, mean

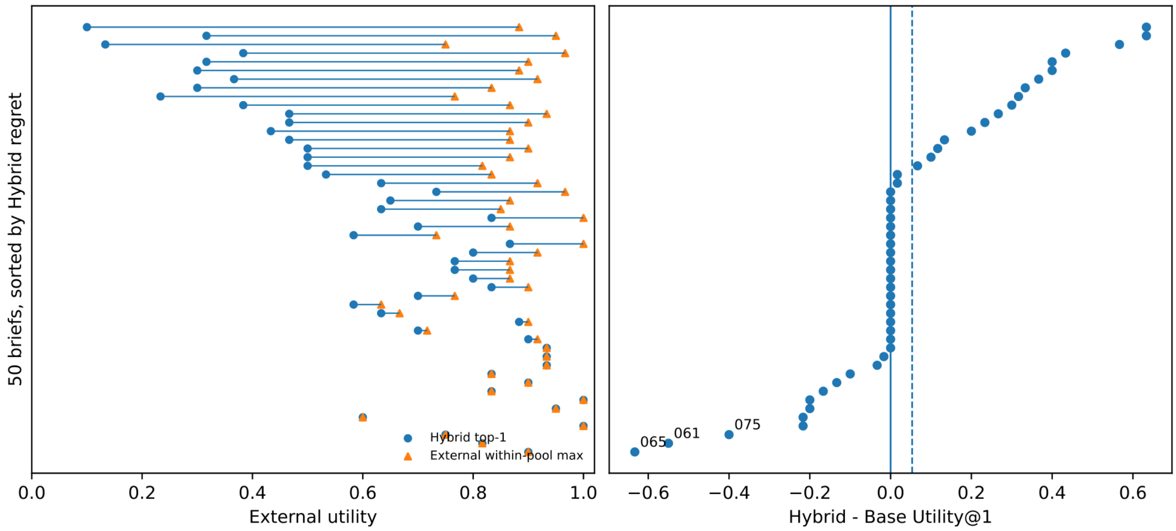

Figure 5: Two views of terminal selection on identical frozen A5 pools. Left: Hybrid top-1 utility and the external within-pool maximum, with briefs sorted by Hybrid regret; horizontal segments show the measured headroom already present in the generated set. Right: Hybrid-minus-Base Utility@1 by brief; the dashed line is the mean +0.0533 difference, which is not statistically resolved. Briefs 061, 065, and 075 are the three largest regressions and are examined in Section 7.3.

parent-to-Focused distance is 0.1834 and parent-to-Structural distance is 0.2409. The paired Structural-minus-Focused difference is +0.0574 (95% CI [+0.0429, +0.0734]); Structural outputs are farther from their parents on 45 of 50 briefs. This is consistent with the transformation permissions defined in Section 4. It does not establish that larger moves are artistically better or that the prompts instantiate formally disjoint operators.

6.3 RQ3: Selection Leaves Substantial Recognition Headroom; the Hybrid Advantage Is Uncertain

Two findings should be distinguished. First, the frozen pools contain a sizeable measured recognition gap. Averaged over briefs, the externally highest-utility candidate already present within the five-proposal A5 set has utility 0.8660. Hybrid surfaces proposals with mean Utility@1 0.6417, leaving mean Regret@1 0.2243. It selects an externally highest-utility candidate first on 13 of 50 briefs (26%), and on 15 briefs no externally highest-utility candidate appears even in its top three. These values concern an external model-based diagnostic, not an objective claim about which artwork an artist should choose.

Second, the experiment does not establish that Hybrid reliably solves this problem better than Base. Hybrid has favorable point estimates on Utility@1, regret, exact top-1, top-2 coverage, and NDCG. Its primary Utility@1 advantage is +0.0533, however, with 95% CI [-0.0160, +0.1240] and paired randomization p = .151. Hybrid yields higher Utility@1 on 19 briefs, the same utility on 19, and lower utility on 12. The appropriate RQ3 conclusion is therefore not selector superiority: selection is empirically separable from search and remains imperfect, while the present tournament policy shows a favorable but statistically unresolved advantage over the simpler Base selector.

Table 17: Terminal-selection metrics on identical frozen A5 proposal sets.

Metric	Base	Hybrid
Utility@1 ↑	0.5883	**0.6417**
Regret@1 ↓	0.2777	**0.2243**
Exact top-1 accuracy ↑	0.18	**0.26**
External maximum in top-2 ↑	0.44	**0.48**
External maximum in top-3 ↑	0.70	0.70
NDCG@5 ↑	0.8758	**0.8954**

6.4 Secondary Selected-Output Check

Complete-pool evaluation is the primary search endpoint because it observes all 25 cross-pool relationships for each brief. A separate blind campaign asks a more conventional end-to-end question: if each condition is collapsed to the single proposal it surfaces first, which selected proposal receives higher holistic preference? Across the 50 held-out briefs, pooled point estimates favor Hybrid-selected A5 in all three contrasts (Table 18).

Table 18: Secondary selected-output holistic comparison over 50 held-out briefs. Values are pooled descriptive preference point estimates for Hybrid-selected A5. Brief-level uncertainty is not reconstructed because the complete per-brief records required for that analysis are unavailable for part of this secondary campaign.

Selected-output comparison	A5 preference
A5 vs. selected B5-self	0.630
A5 vs. selected M5	0.620
A5 vs. selected B5-raw	0.670

We treat these values as a descriptive secondary consistency check and make no separate inferential claim from them here. We therefore retain this campaign as a descriptive directional check and do not reconstruct brief-level uncertainty from incomplete per-brief records. Each contrast observes only one selected pair per brief and reintroduces dependence on internal ranking; the exhaustive pool analysis therefore remains the stronger evidence for RQ1.

6.5 Robustness of the External Model-Based Measurement

The preceding effects are properties of a model-mediated holistic evaluation. The analyses below ask whether the central A5–M5 result is readily explained by several observable features of that measurement process. They do not establish equivalence to professional artistic or curatorial judgment.

Across all three complete-pool campaigns, exact A/B orientations agree on 79.41% of the 3,750 logical candidate pairs. When both orientations report confidence 4–5, agreement rises to 94.73% across 1,803 pairs. The global preference for whichever proposal happens to occupy physical Candidate-A position is 0.5016, close to positional parity. These diagnostics make a simple global position-bias account difficult to sustain, while leaving subtler evaluator biases possible.

For the central A5–M5 comparison, 962 of 1,250 logical pairs are swap-consistent. Restricting the analysis to this different, more stable subset yields A5 preference 0.5608 and a brief-balanced centered effect of +0.1280 (95% CI [+0.0639, +0.1916], randomization $p = .00024$; brief W/T/L 31/10/9). Consensus filtering changes the estimand and can preferentially retain easier comparisons, so this does not replace the full-matrix result. Its narrower implication is that the positive matched-control direction does not depend on comparisons whose verdict reverses under exact presentation inversion.

A5 dossiers are also longer on average than M5 dossiers (469.3 vs 425.1 words). Yet relative length is only weakly associated with pairwise outcome (Spearman $\rho = 0.045$, brief-cluster 95% CI [-0.040, +0.129]), and A5 preference remains above 0.5 when A5 is no longer than its M5 opponent (0.548) and when it is strictly shorter (0.555). These diagnostics make a simple monotonic preference for longer dossiers difficult to sustain without ruling out subtler stylistic effects.

Finally, removing each held-out brief in turn leaves the A5–M5 point estimate positive, ranging from +0.0763 to +0.0959. Removing any complete ten-brief coverage category also preserves the positive direction, with re-estimated effects from +0.0740 to +0.0955. These are sensitivity ranges rather than confidence intervals or category-level hypothesis tests.

Taken together, these checks make several narrow procedural explanations difficult to sustain: a global physical-position preference, dependence on visibly order-unstable pairs, one influential brief, one coverage category, or a simple monotonic preference for longer dossiers. They do not validate the underlying artistic construct. The external evaluator is a separate holistic model-based preference instrument, and stability of its judgments does not establish that artists, curators, institutions, or audiences would make the same comparisons, or that textual proposal differences would survive fabrication.

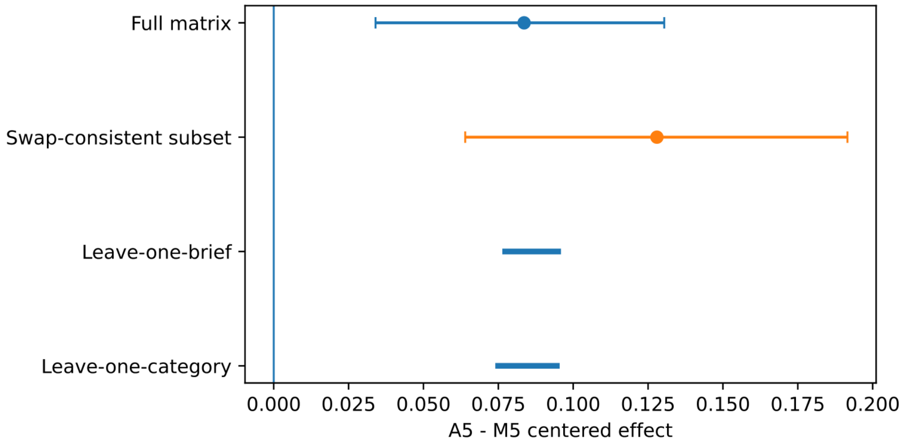

Figure 6: Sensitivity of the central A5-M5 effect. Full and swap-consistent rows show point estimates with 95% bootstrap intervals. Leave-one-brief and leave-one-category rows show only the range of re-estimated point effects after omission; they are not confidence intervals.

7 Mechanism and Failure Analysis

Quantitative effects do not by themselves show what selected transformations change or where failures enter the pipeline. We therefore inspect frozen lineage and ranking traces selected for explanatory contrast rather than statistical representativeness; they add no independent observations to RQ1–RQ3.

(a) Parent — Echoing Gutter

(b) Selected Structural — Layered Ledger

Figure 7: Post-hoc matched renderings of Brief 053. Left: parent Echoing Gutter, in which public repair returns the visible mesh toward reset. Right: selected Structural challenger Layered Ledger, in which repair restores the front layer while repeated pigment transfer leaves an accumulating record on the backing layer. The renderings visualize the frozen textual proposals and were not shown to the evaluator; the external evidence is the blind exact-parent text comparison.

7.1 What Selected Transformations Change

Section 2 introduced the scaffold stakes →artistic operation →realization. The lineage traces make this distinction concrete. A useful intervention need not invent a new topic. It may instead change the relation through which the same pressure becomes operative, shift where causality resides, reorganize the spectator's spatial relation, or make a production constraint constitutive rather than compensatory.

7.1.1 Brief 053: repair stops erasing its own history

Brief 053 asks for contested maintenance using canvas, galvanized mesh, and hand-mixed pigment, with every component repairable in public during opening hours. The parent Echoing Gutter already has a visible damage-and-repair loop: pigmented water stains a galvanized-mesh curtain; a visitor brushes the streak away; the mesh returns toward a uniform state. The weakness identified in the opportunity state is unusually concrete. The repair is fully reversible: every cycle returns the visible work toward its initial condition. Canvas is also peripheral despite being one of the specified materials. The proposed transformation is therefore not "add more evidence of maintenance," but to make repair both restorative and accumulative.

The selected Structural challenger, Layered Ledger, backs the mesh with canvas. A pigment line again drips across the mesh and is again brushed away. This time, however, brushing transfers pigment through the mesh and permanently stains the canvas behind it. The front surface is restored while a ghost trace accumulates on the backing. The operative difference is compact:

Parent	damage $\rightarrow$ public repair $\rightarrow$ visible reset
Challenger	damage $\rightarrow$ public repair $\rightarrow$ front surface restored + history accumulated behind

The selected challenger receives an exact-parent score of 1.0: the external evaluator prefers it in both exact A/B orientations. The complete Brief-053 A5–M5 effect is +0.44. The latter is a property of all five proposals and is not evidence attributable to this one lineage; the case is used because the transformation itself is unusually easy to inspect.

7.1.2 Material causality: Brief 030

Brief 030 provides a different type of transformation. Chronicle of Pull makes inherited obligation uneven through a shared veil pulled by weighted cables whose behavior is mediated by frozen clocks and differing mechanical resistance. The selected Structural challenger Weight of Inheritance keeps the shared veil and unequal response but removes clock mediation: reclaimed brick masses of unequal weight themselves become the source of unequal pull. The transformation therefore relocates causal emphasis. The parent uses a temporal apparatus to mediate differential burden; the challenger makes differential material burden directly responsible for the shared surface's unequal movement. The selected challenger wins both external orientations against its exact parent. This is not evidence that every successful Structural intervention behaves this way; it is a concrete example of a material-causal reorganization.

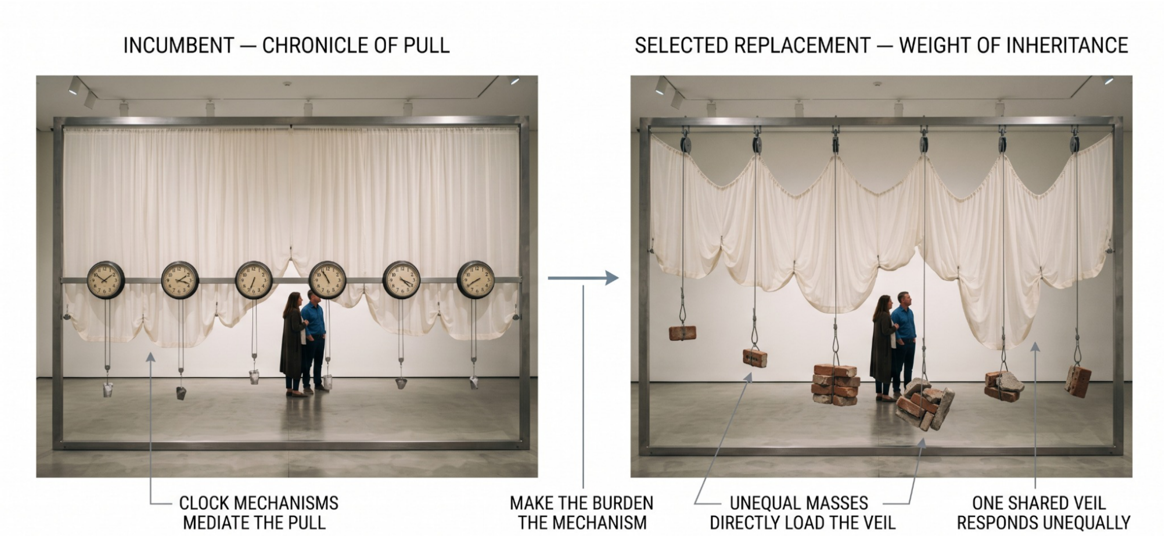

Figure 8: Post-hoc visualization of a selected Structural transformation in Brief 030: Chronicle of Pull →Weight of Inheritance. The shared veil and unequal response remain, while clock-mediated pull is replaced by unequal physical masses, moving the causal burden into the material itself. Renderings were produced after evaluation and provide no independent quality evidence.

7.1.3 Spatial encounter: Brief 038

Brief 038 constrains the work to chalk, reclaimed brick, and rainwater-safe sealant and asks that a disputed boundary remain legible under partial weathering. Eroding Chalk Line Labyrinth organizes the boundary primarily as a linear or frontal condition. The selected Structural challenger Fragmented Borders: Erosion Field distributes reclaimed bricks as an irregular walkable lattice and carries the chalk boundary through that field. The core materials and the problem of differential weathering remain recognizable while the spectator relation changes: boundary encountered as demarcation →boundary negotiated as terrain. The selected challenger wins both exact-parent orientations.

Yet the complete Brief-038 A5 set performs substantially worse than M5 ($\Delta U = -0.52$). This mismatch becomes useful in Section 7.2: a successful local intervention does not imply that the five-proposal search succeeded as a whole.

Figure 9: Post-hoc visualization of the selected Structural transformation in Brief 038: Eroding Chalk Line Labyrinth → Fragmented Borders: Erosion Field. A boundary encountered frontally becomes a field through which visitors move. The visualization supports interpretation of the textual change and is not independent evidence of quality.

7.1.4 Institutional constraint as operative structure: Brief 059

Brief 059 asks for a university-commons installation about interrupted scholarship that can be paused for events, stored in labeled cases, and restored by non-specialist staff from a one-page diagram. The parent Retractable Lecture Shelf makes suspension visible by lowering an essay-bearing steel shelf into a recessed floor cavity with a motorized lift. The artistic premise is coherent, but the realization fits the explicit storage requirement awkwardly and depends on a bespoke built-in mechanism. The selected Focused reconstruction, Fold-Out Fabric Archive, preserves the visible suspension of scholarship while replacing the motorized track with a low-tech folding structure. Staff manually fold the work into labeled cases and later reconstruct it from the diagram. The institutional action required by the brief—pause, case, restore—therefore becomes part of the work's own visible operation rather than an accommodation appended after the concept. The selected replacement wins both external orientations against its exact parent.

7.2 Contestable Refinement Can Succeed Locally, Fail Locally, or Leave the Parent Untouched

The strong mean RQ2 result can obscure an important fact: the 101 adopted interventions are not uniformly successful. Seventeen swap-consistent selected replacements lose to their exact parents. Brief 038 makes the resulting hierarchy of outcomes especially visible because several different things happen inside the same five-lineage search. This brief is useful precisely because it resists a success-story reading. A coherent local improvement in one lineage neither guarantees that other interventions in the same brief are useful nor that the overall search policy constructs a stronger five-proposal set than another policy. RQ2 measures the consequence of a selected intervention relative to its exact parent; RQ1 evaluates the resulting candidate population as a whole.

Brief 050 supplies a cleaner single-lineage failure. The brief asks for an interrupted secular ritual without familiar religious or collective-healing vocabulary. The incumbent Threaded Pause stages a shared weaving process whose continuity is abruptly broken by a timed cutter. The opportunity state identifies a plausible risk that the cut may become a one-off gimmick, but its proposed response is itself questionable. The selected Structural challenger Resonant

Pause keeps the cutter and adds a short white-noise burst and light cue around the rupture. The external exact-parent outcome is unambiguous: the selected challenger scores 0.0, meaning that the parent is preferred in both swapped orientations. This trace demonstrates the limited status of intermediate reasoning. A defensible diagnosis can still yield a proposed repair that the downstream external evaluator prefers less than the original state. Reviews and opportunity states therefore function as search evidence, not certificates of improvement.

Table 19: Brief 038: local intervention outcomes within one five-lineage search. Exact-parent outcome is derived from the two swapped external comparisons.

Lane	Final state	Exact-parent outcome
1	Focused reconstruction	loss in both orientations
2	Structural escape	win in both orientations
3	Focused reconstruction	loss in both orientations
4	Incumbent retained	identity tie
5	Incumbent retained	identity tie
Complete five-proposal A5 vs M5 effect: **-0.52**		

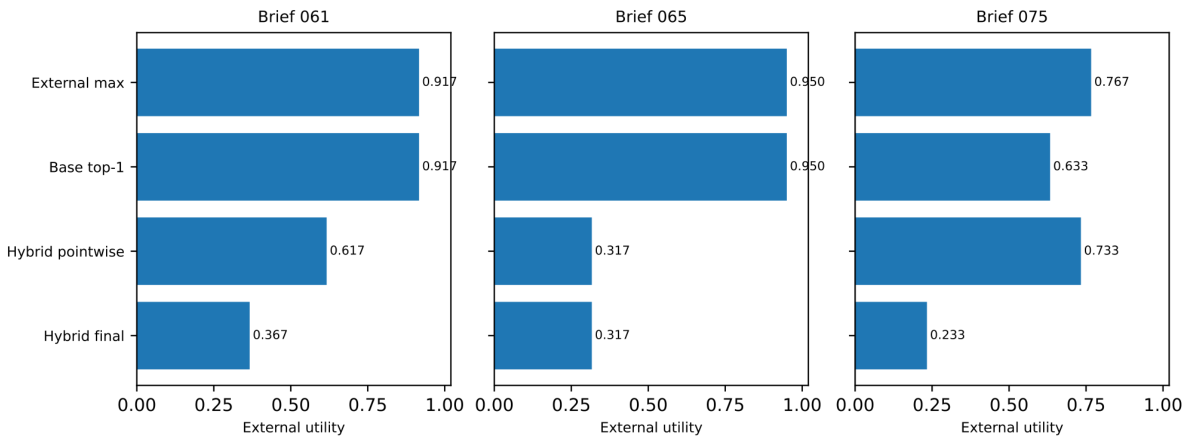

Figure 10: Three locations of terminal-selection error. Brief 061 loses the external maximum before tournament comparison; Brief 065 retains it among finalists but fails to recover it; Brief 075 regresses because tournament aggregation overturns a better pointwise choice. External utility is a diagnostic measured after generation, not an internal ranker score or a claim of objective artistic value.

7.3 Terminal Selection Errors Arise at Different Stages

The RQ3 aggregate identifies recognition headroom but does not reveal whether errors arise during broad pointwise screening or during the finalist tournament. A post-hoc stage-specific diagnostic first guards against cherry-picking individual tournament failures. Relative to the Hybrid pointwise top choice, the final tournament changes top-1 on 20 of 50 briefs. Its mean external Utility@1 change is +0.0150, with 95% bootstrap interval [−0.0413, +0.0747]; brief W/T/L is 9/30/11 and a paired randomization test is not significant ($p = .631$). Thus the experiment does not establish that tournament comparison is beneficial or harmful on average.

Against that aggregate background, the three largest Hybrid-versus-Base regressions are informative because they localize three different failure paths:

• Brief 061 - screening failure. The externally highest-utility candidate, Weightless Ledger (0.917), is ranked outside the Hybrid finalist set. Base surfaces it first; Hybrid ultimately surfaces Morphing Transfer (0.367). The pairwise stage never gets the opportunity to recover the higher-utility candidate.

• Brief 065 - failure of recovery. Accumulating Ink Pool is the externally highest-utility candidate (0.950) and Base ranks it first. Hybrid pointwise ranking places it among the finalists but puts Midway Descent first; the tournament does not correct that ordering and the final top choice remains at 0.317.

• Brief 075 - tournament-induced regression. Hybrid pointwise ranking selects Echo Chamber (0.733), which would outperform the Base top proposal (0.633). Tournament aggregation overturns it and surfaces Heat Ledger (0.233); the external within-pool maximum is higher still at

0.767.

These cases localize distinct selection failures without implying that tournament comparison is harmful on average.

8 Discussion

8.1 What the Decomposition Reveals About Concept Development

For contemporary-art concept development, the central empirical result is not a single end-to-end score but the fact that three consequential stages behave differently under evaluation. A weak surfaced proposal may reflect a weak field of alternatives, an intervention that damaged a stronger state, or a failure to recognize which surviving proposal deserved attention. Once plausible proposals are inexpensive to generate, those failures are easily conflated by final-output evaluation.

The experiment makes these failure modes separately observable by assigning sampled-set quality, exact-parent intervention, and terminal ranking to distinct empirical objects. The evidence is deliberately asymmetric: candidate construction and adopted replacements improve under the external measure, whereas terminal selection remains unresolved.

This asymmetry is useful beyond the domain because one compelling top-ranked output can conceal a weak candidate population or a harmful development policy, while a poor surfaced proposal can hide stronger material that already exists. Treating the sampled set, lineage transition, and terminal ranking as separate empirical objects therefore turns a property made especially visible by installation concept development into a more general diagnostic for creative agents.

The decomposition is operational rather than ontological. We do not claim that creativity consists of three independent modules, nor that these stages correspond literally to how artists develop work. Their value is experimental: they expose failure modes that an end-to-end winner-only evaluation would otherwise merge.

8.2 Search Semantics Matter in Addition to Search Width

The matched M5 comparison provides the strongest evidence that the organization of concept development matters beyond additional generic iteration. M5 starts from the same self-refined parents and preserves much of the same iterative scaffold, yet A5 retains a +0.0836 complete-pool effect. Computationally, this is evidence about local-search semantics: AIrtist converts critique into an opportunity state that distinguishes what should be preserved, what most needs intervention, which field attractor to escape, and which unused degree of freedom to exploit, then gives Focused and Structural challengers different permissions around that state.

The post-hoc operator analysis provides a behavioral consistency check on this interpretation. Structural candidates occupy farther parent-relative semantic neighborhoods than Focused candidates on 45 of 50 briefs under the chosen representation. This does not establish that operator heterogeneity causes the A5-M5 gain, nor that larger transformations are better. It does show that the two instructions are not merely different labels attached to indistinguishable rewrites: the intended difference in transformation scope is visible in their outputs.

The causal boundary remains important. M5 is substantially matched rather than token- or call-identical, and several elements of the domain-structured search package change together: specialized reviews, opportunity synthesis, preservation and forbidden-fix constraints, field differentiation, and challenger roles. The present experiment therefore cannot identify which element is necessary, which is redundant, or whether several operate synergistically. The next step for mechanism attribution is ablation, not a stronger claim about any one prompt component.

More broadly, the result suggests a useful axis for inference-time search in open-ended domains. Search is often described through depth, branching factor, number of samples, critique rounds, or voting procedures. The present evidence indicates that the semantics assigned to a branch can matter alongside the amount of branching. Where revisions trade one virtue for another and no verified scalar reward is available, specifying what a move is allowed to change - and what it must protect - may be as consequential as allocating additional inference to the move itself. Whether this transfers beyond installation ideation remains an empirical question.

8.3 Additional Inference Can Test Whether to Change

Competitive refinement suggests a different use of additional inference: proposed revisions can be tested without making revision compulsory. Conditional on replacement, the frozen policy tends to adopt interventions preferred to the exact states they displace.

This distinction suggests a different role for additional inference in open-ended systems. Computation can be spent testing whether a current state deserves intervention, not merely searching for a successor that looks more developed. That design principle is potentially relevant wherever revision can exchange rather than monotonically accumulate virtues - for example in design, writing, planning, or research ideation - but the present experiment establishes it only for this proposal-development setting.

Nor does the experiment establish that optional retention is causally responsible for the parent-relative gain. A matched mandatory-replacement policy was not generated. Such a condition would be required to determine whether allowing the incumbent to survive is itself better than always choosing the strongest available challenger.

8.4 Better Measured Candidate Sets Need Not Be Broader Candidate Sets

The post-hoc breadth analysis shows that multiplicity does not imply broader exploration: under the chosen latent-semantic representation, A5 is more concentrated than M5 despite its higher measured proposal quality.

This matters because sampled-set quality and breadth answer different questions. The quality analysis asks whether the retained proposals fare better under the external holistic evaluation. The breadth analysis asks how far apart proposals lie under one particular semantic representation. A system can improve one without improving the other. The present result therefore argues against using the existence of several agents, lanes, roles, or outputs as evidence that a system has preserved diversity.

The result should also not be inverted into a new optimization principle. It does not show that concentration causes higher quality, that narrower pools are preferable, or that A5 is less diverse in every meaningful artistic sense. The rank association between quality gain and breadth difference is exploratory, and latent-semantic distance captures only one notion of difference. Two proposals can be textually close while differing importantly in fabrication or institutional meaning; they can also be semantically distant while remaining variants of the same familiar artistic operation.

A more useful design implication is that quality and breadth may need separate controls. An open-ended creative system may want to retain several proposals above a quality threshold while explicitly preserving differences in operation, material strategy, or spectator relation, rather than assuming that a single search objective will maximize both. This also suggests a stricter future evaluation: breadth should be measured directly and at several representational levels instead of inferred from branch count or agent multiplicity.

8.5 Selection Is Attention Allocation, and the Recognition Problem Remains Unresolved

The selector experiment establishes the recognition problem more clearly than the present solution: stronger measured material often already exists in the pool, while Hybrid's advantage over Base remains statistically unresolved.

For an artist receiving five developed dossiers, ranking is first an allocation of attention over already available material: it shapes which possibility is encountered first, which comparison anchors the rest of the set, and which generated directions may never receive serious attention. The same structural problem extends to any creative agent that produces more alternatives than a user will inspect equally, whether or not the system calls the policy a ranker.

The present data support separating this problem from generation. They do not establish the optimal selector. Future work should examine ranking objectives, uncertainty-aware presentation, and perhaps policies that preserve incomparability rather than forcing every candidate set into a total order.

8.6 Domain Grounding Is Both an Inspectable Prior and an Inspectable Error Surface

AIrtist's domain grounding encodes a recognizable position. Its concept-development policy privileges enacted artistic operation over topical illustration, material and spatial necessity over arbitrary styling, specific spectator relations over generic participation, field differentiation over first-obvious operating formats, and structural repair over additive explanation or spectacle. These commitments enter through the internal quality model, specialized reviews, opportunity representation, and challenger instructions.

The A5-M5 result suggests that making such search priors explicit can be useful. It is preferable experimentally to an opaque instruction such as "make this more creative," because an explicit prior can be inspected, criticized, edited, and ablated. The system exposes not only a proposed improvement but the terms under which it sought that improvement: what it attempted to preserve, what it treated as the dominant liability, what neighboring format it tried to escape, and what class of repair it refused.

The failure cases show the other side of the same decision. Domain grounding does not remove the error surface; it reshapes it. A search pressure designed to escape familiar tropes can over-transform a direct solution. A demand for material specificity can encourage additional apparatus. A plausible critique can motivate a repair whose added mechanism is weaker than the parent. Brief 038 is particularly instructive: one Structural intervention is locally successful against its exact parent, while two other adopted interventions fail and the complete A5 pool performs substantially worse than M5. The same structured machinery that helps the search on average can therefore push individual lineages past a stronger economical state.

This is important for how domain expertise should be represented in creative agents. A domain-specific prior should not be treated as expert truth added to a generic model. It is better understood as a contestable search position that changes which errors become likely. Making that position explicit creates two advantages: it can improve search under the conditions studied here, and its characteristic failures become available for diagnosis rather than disappearing inside an undifferentiated "creative" prompt.

Contemporary-art installation ideation makes these issues unusually visible because three properties that can remain hidden in more easily scored domains are constitutive here: revision is non-monotonic; relevant qualities are plural and can conflict; and material, spatial, spectator, and institutional relations cannot be reduced to surface correctness. The domain therefore functions not merely as an application area but as a stress test for creative-agent design under weakly specified reward.

The experiments support a correspondingly bounded backend principle: maintain several viable states long enough to compare them, make intervention conditional rather than compulsory, represent the intended transformation explicitly, and audit selection separately from generation. They do not establish how an artist will use those states in practice.

9 Intended Artist-Facing Use: Idea Search as a Substrate for Negotiated Co-Creativity

The preceding discussion established what the autonomous concept-development backend does computationally; this section asks how those properties could become negotiable to an artist. The intended use begins when an artist has a direction but not yet a specification: they may know what matters, which conditions or materials constrain the problem, and which obvious solutions to exclude while remaining uncertain about what the work should become. AIrtist turns one such rough brief into five ranked project hypotheses. The experiment measures that computational substrate; negotiation over its assumptions and outputs remains a subsequent interaction problem.

9.1 From a Provisional Query to Five Project Hypotheses

A conventional generative interface often treats the prompt as a specification: the user states what is wanted, and the model attempts to produce an answer. This model is poorly matched to early creative work, where part of the difficulty is precisely that the desired answer is not yet known.

As an information-retrieval analogy, we treat the artist's brief as a provisional query into a much larger space of possible projects. The analogy is useful because the brief identifies a local problem without exhausting its solution: its free-text premise states what is at stake; must-address conditions protect pressures the artist does not want development to lose; must-avoid conditions eliminate directions already known to be unwanted; and site, material, institutional, or production requirements further constrain the region to be explored. The resulting input specifies an under-determined development problem rather than a target artifact.

The analogy should not be read literally as document search. AIrtist does not retrieve an existing artwork or query a database of pre-authored concepts. Its computational mechanism is generative search over proposal states: candidate project directions are constructed, reviewed, transformed, preserved or displaced, and finally ranked. What the artist receives is therefore not a list of retrieved answers but a locally constructed field of possible responses to the brief.

In the evaluated configuration, search after the brief is autonomous and returns five comparatively developed proposal dossiers rather than short brainstorming fragments. Their artistic operation, spatial and material logic, spectator relation, and production implications are explicit as described in Section 4. The dossiers are starting directions for subsequent artistic work, not five artworks from which the model has identified the correct one: rank 1 marks where AIrtist allocates attention first, not what the artist ought to make. Human artistic agency resumes when the resulting possibility set is returned.

9.2 The Sampled Possibility Set as a Structured Field of Alternatives

Treating the returned proposals as a field of alternatives connects the computational object studied in the experiment to a broader search-and-negotiation account of co-creativity. If a collaborator already knew the exact desired solution, one optimized completion might be sufficient. The more characteristic early-stage case is different: the collaborator needs help making the alternatives themselves concrete enough to think with. The computational role of search is then not simply to maximize the probability of one apparently optimal answer, but to construct a useful field of candidate directions for subsequent human judgment.

RQ1 supports treating this field as a first-class output: relative to matched M5, the measured gain extends beyond one oracle-best candidate into the pool mean, upper three proposals, and lower tail. The practical point is not to preserve all five indefinitely, but to avoid treating the surrounding alternatives as disposable scaffolding.

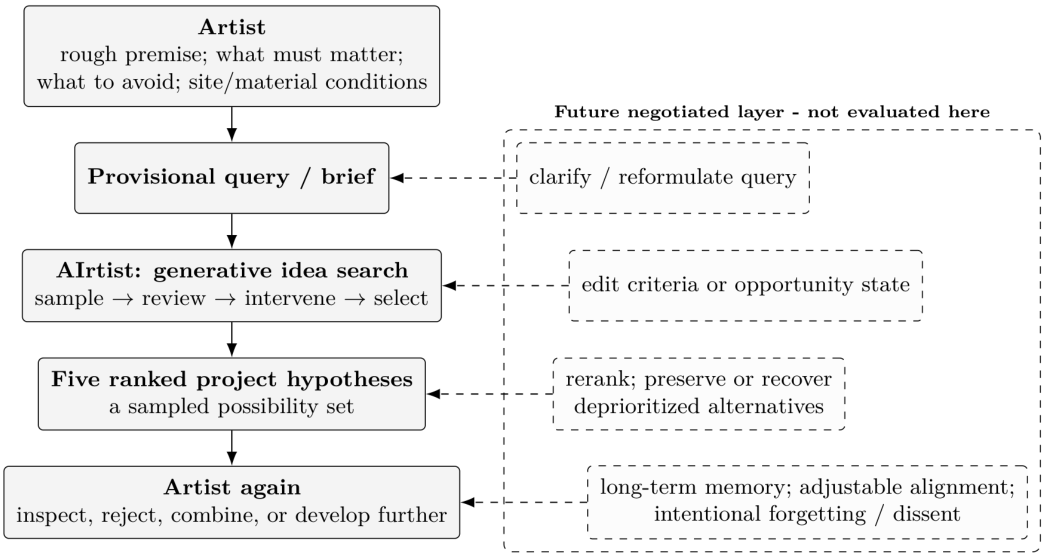

Figure 11: From generative idea search to negotiated co-creativity. In the evaluated system, an artist-authored provisional brief is autonomously expanded into a ranked sampled possibility set of five complete project hypotheses. There is no artist feedback inside the evaluated pipeline. Clarifying questions, editable search states, criteria negotiation, branch recovery, and longitudinal personalization are future interaction layers rather than capabilities evaluated in the present study.

The post-hoc breadth result adds a qualification: several proposals do not automatically constitute broad exploration, and branch or agent count cannot stand in for diversity.

This points toward a richer model of generative idea search. Quality and breadth are potentially separate search objectives. A future system might explicitly diversify candidate operations, cluster related proposals, preserve alternatives from different regions, or maintain a small number of minority directions even when they are not favored by the dominant ranking criterion. The present AIrtist configuration does not implement such a diversity-aware candidate policy; the current result instead motivates testing such policies explicitly.

More generally, a structured field of alternatives need not mean maximum dispersion. Two distant proposals can differ superficially while relying on the same artistic operation, whereas relatively close proposals can expose consequential differences in material causality, institutional procedure, or spectator position. For an artist-facing search system, the relevant problem is therefore not simply to return the most dissimilar five items, but to make several consequentially different directions available without allowing low-quality variation to consume the entire field.

9.3 Ranking as Local Visibility and Attention Allocation

Generating several alternatives creates a second problem familiar from information retrieval: not everything that exists in the candidate collection will receive equal attention.

Ranking is therefore not merely a presentation convenience. Within the sampled possibility set, it determines local visibility: which generated direction is encountered first, which proposal anchors comparison with the others, and which candidates risk being overlooked. At larger cultural and institutional scales, ranking systems can govern visibility in much broader ways. The present experiment makes no such culture-wide claim; it studies the smallest instance of the same structural problem, namely visibility among five candidate project hypotheses produced for one artist-authored brief.

RQ3 supplies the empirical reason to keep ranking provisional: higher-utility measured material often already exists in the generated pool without being surfaced first, while the present Hybrid advantage over Base is unresolved.

An artist-facing interpretation should preserve both findings. Ranking offers one way to manage the non-trivial attention cost of five developed dossiers. But the ordering should remain provisional rather than authoritative. A concise way to state the intended semantics is:

The ranking says which proposal AIrtist recommends reading first, not which proposal the artist should make.

This is also why the intended output remains the ranked set rather than rank 1 alone. Suppressing lower-ranked candidates would give the selector more authority precisely where the experiment reveals considerable remaining recognition error. Returning all five keeps deprioritized options recoverable and allows the artist to disagree with the system's allocation of attention.

This creates a direct connection between information retrieval and co-creative system design. Relevance is no longer only a property of documents retrieved for an information need; it can govern which generated possibility becomes creatively visible within an interaction. Future systems could expose different orderings of the same candidate field according to different criteria—for example conceptual necessity, novelty, feasibility, public or ethical risk, or distance from a creator's habitual solutions—rather than collapsing these concerns into one hidden score.

The present experiment intentionally does not do this. One bounded internal working model is frozen so that search policies can be compared without simultaneously changing the criterion of preference. Criteria-contestable ranking is therefore an interaction problem for a subsequent system, not an unreported capability of AIrtist.

9.4 Autonomous Search Is Not Yet Negotiated Agency

The evaluated system implements search, but not yet the second half of a broader search-and-negotiation model of human–AI co-creativity.

This boundary is substantive. AIrtist does not ask the artist clarifying questions when the brief is underspecified. It does not show the artist its opportunity memo before generating challengers. The artist cannot alter what the system has decided to preserve, reject its diagnosis of the dominant liability, change the field logic it attempts to escape, unlock a prohibited transformation, or ask it to recover a deprioritized branch while the run is in progress. Nor does the current system adapt its quality criteria to an individual artist.

Competitive refinement nevertheless creates some of the computational objects around which such negotiation could later occur. Inside the system, intervention is already contestable in a limited sense: an incumbent remains eligible and a proposed change must earn replacement. Human contestability would add a different level of agency. The collaborator would be able to contest not only the generated candidate but the policy that produced it.

Competitive refinement makes intervention contestable inside the computational search; negotiated co-creativity would make the search itself contestable to the human collaborator.

The opportunity state is particularly useful for negotiation because it externalizes otherwise tacit steering decisions: what the system intends to preserve, change, escape, exploit, and refuse as an adequate repair.

In the present experiment these fields are internal search state. In a negotiated system they could become objects of disagreement. An artist could agree with the diagnosis but reject the proposed escape; preserve a mechanism the model wants to remove; declare that an inefficient material process is essential; or deliberately permit a solution the default quality model treats as too didactic, decorative, spectacular, or familiar.

This would change the role of AI initiative. A proactive system could ask questions, point out contradictions, propose transformations, criticize a direction, or refuse an unsafe implementation. But each of these actions also risks allowing the system to define the artistic problem on the artist's behalf. Negotiated agency therefore requires more than additional conversational turns. It requires making visible when the system is assisting, when it is steering, and what assumptions authorize that steering.

9.5 From Generative Search to Negotiation and Long-Term Collaboration

The present architecture suggests a natural sequence for extending autonomous idea search into negotiated co-creativity.

The first extension is query negotiation. If the initial brief is a provisional query rather than a complete specification, the system need not assume that the first formulation is sufficient. It could detect unresolved constraints, tensions between must-address conditions, or ambiguity about what is intended to remain open, then ask a small number of targeted questions before committing search resources. The important research question is not simply whether clarification improves proposal scores, but when clarification helps articulate an artist's intention and when it instead prematurely determines the problem for them.

The second extension is search-state negotiation. Opportunity states could be exposed before transformation. Artists could edit or lock individual fields, request alternative diagnoses, or ask different agents to formulate competing readings of the same incumbent. A disagreement over what should be preserved could then generate genuinely different searches rather than another rewrite under one hidden interpretation.

The third extension is recoverable selection. Ranking need not permanently erase losing candidates. A future interface could preserve minority alternatives, permit criterion-specific reranking, display proposals whose pairwise ordering is unstable, and allow the artist to recover a branch that the model deprioritized. The current recognition gap makes this especially important: if selection is fallible, recoverability is not only an interface convenience but a safeguard against premature collapse of the candidate field.

These extensions move AIrtist toward negotiated agency without requiring that every internal model operation become a conversational turn. Human involvement can be selective. An artist might delegate broad initial search, intervene only when the system identifies a consequential conflict, and reclaim control when a ranking decision would otherwise remove an interesting direction from view.

A further extension is longitudinal. The current AIrtist has no persistent model of an individual artist. Every user receives the same frozen search machinery and bounded working criteria. Long-term creative collaboration, however, creates an obvious role for memory. Retrieval and personalization could make recurring concerns, materials, rejected clichés, philosophical commitments, past projects, and unresolved ideas available to later searches.

Such memory also creates a distinctive failure mode: aesthetic enclosure. A system that becomes increasingly good at predicting what an artist has previously accepted may progressively reduce the chance of encountering what the artist has not yet learned to want. Personalization can therefore become another form of fixation.

The present results make this a concrete hypothesis for longitudinal testing. A5 already shows that higher measured quality need not coincide with greater latent-semantic breadth. A longitudinally personalized agent should therefore not assume that increasing fit to an artist's prior preferences will preserve productive difference. Future systems may need mechanisms explicitly designed to counter enclosure: different temporal layers of memory, intentional forgetting, adjustable degrees of alignment, or separate agents in which one remains closely aligned while others deliberately retrieve or generate dissenting and unfamiliar directions.

These controls would also have to make creative data governance visible. An artist should be able to inspect what personal material is being treated as precedent, decide what may be retained or forgotten, and distinguish persistent artistic commitments from temporary experiments that should not harden into a permanent profile.

The boundary is intentional: this paper evaluates the search substrate, not whether artists find the resulting field generative, whether negotiation changes their intentions, or whether longitudinal memory supports rather than encloses an evolving practice. Those questions require interactive and longitudinal studies.

10 Limitations and Threats to Validity

10.1 Construct Validity: Model Preference Is Not Artistic or Curatorial Judgment

The dominant limitation is construct validity. The principal outcomes in this study are derived from blind holistic comparisons performed by GPT-5.6 Sol. Exact order reversal, consensus analysis, verbosity diagnostics, and leave-out tests probe the stability of this measurement procedure. None establishes that model preferences coincide with those of professional artists, curators, producers, institutions, or publics.

This boundary is particularly important because two model-mediated evaluative structures appear in the system and should not be conflated. AIrtist's internal ten-dimensional working model shapes review and ranking through explicit concerns such as prompt fidelity, enacted tension, artistic-operation necessity, field distinctiveness, spatial and visual encounter, material necessity, spectator specificity, reception resilience, and production/institutional viability. The external evaluator, by contrast, performs a separate holistic A/B judgment of complete proposals. It is instructed to compare the artistic proposition as a whole rather than mechanically reproduce the internal scorecard. The two instruments overlap in the concerns they attend to, but they are not the same measurement.

Consequently, the main results establish comparative preference under a disclosed model-based evaluation instrument; they do not validate the internal quality model as a general theory of contemporary art. Nor does separating the generator from the external evaluator solve the larger construct-validity problem. The separation is important experimentally because external utility is unavailable to the generator, lane selector, and terminal rankers, but the resulting preferences remain model-mediated rather than professionally validated artistic judgments.

The internal working model itself is intentionally normative. It favors operational necessity, material and spatial specificity, explicit spectator relations, field differentiation, resistance to generic additive spectacle, reception resilience, and plausible institutional realization. These are defensible pressures for the constrained installation-ideation problem studied here, but they are neither neutral nor exhaustive. Practices organized around decorative excess, explicit pedagogy, social service, narrative directness, spectacular display, devotional function, deliberate impracticality, or institutional antagonism may be valued differently by other artistic and curatorial positions. Making these commitments explicit improves inspectability; it does not turn them into universal criteria of artistic quality.

The same limitation applies to model-generated reasons. Judge rationales are useful in the Mechanism and Failure Analysis because they reveal what the evaluator says it attended to—for example, a material violation, excessive apparatus, or an unclear spectator relation. They are not an independent source of expert interpretation and should not be treated as causal explanations of why one proposal is artistically superior. Verdict and explanation originate from the same model-mediated instrument.

The strongest claim the study therefore supports is intentionally narrower than "AIrtist makes better art": under a frozen experimental setting and external holistic model evaluator, its candidate sets and selected lineage interventions exhibit the measured differences reported in Section 6. Establishing correspondence with professional artistic judgment requires a separate human validation study.

10.2 External and Ecological Validity: A Curated Benchmark of Unrealized Proposals

The study evaluates 50 held-out commission-style briefs, IDs 026–075, disjoint from the 25 briefs available during system development. The main matched-control direction survives removal of every individual brief and every complete coverage category. This supports the aggregate comparison on the held-out benchmark, but it should not be confused with evidence from a natural sample of contemporary-art commissions.

The briefs were constructed within the project to cover five deliberately balanced forms of pressure: thematic/conceptual, site-specific, material-constrained, institutional/production-constrained, and anti-cliche/difficult. These categories are useful for coverage and sensitivity analysis, but they are not an empirical taxonomy of contemporary-art practice, and their equal frequency is artificial. The experiment can therefore support aggregate claims about performance on this controlled set of constrained installation problems. It cannot estimate how often the same advantage would occur across naturally distributed museum commissions, artist-authored project notes, open calls, gallery briefs, or other cultural contexts.

The benchmark size also places a specific boundary on subgroup interpretation. Each coverage category contains ten briefs. Leave-one-category-out analysis is useful as a robustness check of the overall matched-control direction; it is not evidence that AIrtist performs differently across artistic categories. Likewise, individual success and failure cases should not be interpreted as prevalence estimates for particular artistic mechanisms.

More importantly, the evaluated objects are textual project hypotheses, not realized installations. A dossier can make a coherent case for material behavior, scale, circulation, duration, maintenance, public encounter, staffing, and budget and still fail when produced. No engineering study, conservation review, accessibility review, fabrication process, installation period, maintenance cycle, commissioning decision, or audience observation validates the proposed works.

This text-to-work gap is unusually consequential in installation practice. Sound behaves differently in an actual room; materials wear unpredictably; visitors ignore intended routes; access and fire requirements alter spatial arrangements; natural light changes; repeated interaction modifies the work; and institutional labor that appears minor in prose may become constitutive in practice. Textual proposal evaluation can reason about such relations only counterfactually.

The post-hoc renderings do not close this gap. Their purpose is to make transformations in source dossiers visually legible—for example, a boundary moving from a frontal condition into a walkable field. Image generation can also supply convincing architecture, fabrication detail, lighting, scale, and finish that the text did not earn. Renderings are therefore explanatory visual hypotheses, were never shown to the evaluator, and provide no independent evidence that the represented work would function as depicted.

10.3 Internal Validity and Mechanism Attribution

M5 substantially matches the iterative scaffold but is not compute-identical, and several domain-structured components change together. The A5–M5 result is therefore a package-level comparison; component attribution requires targeted ablations.

RQ2 likewise supports only a conditional claim: adopted replacements are preferred on average to their exact parents, but no matched forced-replacement policy was generated. The experiment therefore cannot show that allowing retention itself causes higher quality or that competitive refinement is superior to mandatory replacement.

RQ3 cleanly isolates ranking because Base and Hybrid receive identical candidate pools, but the observed Hybrid-minus-Base Utility@1 advantage is not statistically resolved. The supported claim is therefore that selection is a separable recognition problem, not that the present tournament is a reliably superior solution.

Finally, all core generation conditions use the same base generation model family. This is an advantage for the within-study comparison because variation in base-model capability does not confound the search-policy contrasts. It leaves model-generalization open. A stronger generator might reduce the marginal value of structured refinement; a weaker generator might benefit more from it; and a differently trained model might respond differently to field-escape or preservation instructions. Cross-model generation experiments are required before treating the architecture as model-independent.

10.4 Evaluation-Procedure Dependence and Exploratory Analyses

The primary external evaluation is exhaustive within each brief but not composed of independent judgments at the pair level. Each logical A5–control pair is presented in exact A/B-reversed orientations, but multiple comparisons within one pass are processed inside the same fresh Temporary Chat. Opposite orientations are isolated in separate chats, making order instability observable, but comparisons within a pass can still be affected by contextual carryover.

For this reason, the study treats the brief rather than the candidate pair as the inferential unit. The 7,500 physical judgments provide dense measurement of 50 held-out tasks; they do not constitute 7,500 independent benchmark observations. A stricter future design could place every logical comparison—or small randomized micro-batches—in an independent context.

Exact swaps expose rather than eliminate evaluator instability. Overall swap agreement is 79.41%, rising to 94.73% among pairs where both orientations report confidence 4–5. Consensus-only analysis shows that the matched-control direction does not depend on the comparisons whose verdict visibly reverses under presentation order, but selecting only stable pairs changes the estimand and may preferentially retain easier distinctions. It should therefore remain a sensitivity analysis rather than be interpreted as a cleaner ground truth.

Relative-length analysis addresses another identifiable threat but should not be overread. A5 dossiers are somewhat longer than M5 dossiers, while the association between relative length and preference is weak and the A5 direction persists when A5 is not longer or is strictly shorter. This makes a simple monotonic verbosity explanation difficult to sustain. It does not establish that the holistic model judge is free of subtler stylistic biases involving fluency, curatorial vocabulary, structural completeness, tone, or apparent professional confidence.

Several analyses in the paper are explicitly post-hoc. Latent-semantic pool breadth, the relationship between quality effect and breadth difference, parent-to-Focused versus parent-to-Structural semantic distance, and the pointwise-to-tournament stage diagnostic were introduced to understand the frozen experiment rather than as members of the planned primary endpoint family. Their role is mechanistic and hypothesis-generating. In particular, lower A5 breadth under one semantic representation should not be generalized to diversity in every artistic sense, and the observed quality–breadth relation does not establish that concentration causes the measured quality gain.

The qualitative cases are selected for explanatory contrast rather than prevalence estimation. They illuminate mechanisms measured elsewhere and add no independent statistical evidence.

AIrtist's field-context construction introduces a further information-quality boundary. The system can use model-mediated recall of artists, works, texts, and neighboring formats to construct a shared field map. This is associative context construction rather than provenance-bearing retrieval. Such recall can be incomplete, inaccurate, culturally narrow, or anachronistic. Recalled precedents are used as internal prompts for differentiation rather than as factual art-historical evidence in the external evaluation. A stronger future system should separate associative recall from verifiable retrieval and expose provenance when factual claims about artistic precedent matter.

10.5 Human Co-Creative Validity Is Outside the Present Experiment

The intended artist-facing use model makes a final boundary especially important. The present study evaluates an autonomous concept-development backend, not negotiated human–AI co-creativity.

An artist supplies a partially specified brief once. AIrtist then generates, reviews, intervenes, and ranks autonomously before returning five project hypotheses. The experiment does not observe whether artists find those hypotheses generative, whether they prefer the structured set to alternatives produced by another system, whether they combine or reject proposals, whether rank position causes fixation, or whether the opportunity representation helps them articulate their own intentions.

Accordingly, improved sampled-set quality should not be rewritten as improved human creativity, creative satisfaction, authorship, trust, agency, or co-creative experience. Likewise, returning several proposals creates the possibility of comparison but does not establish that five is the right number for a working artist, that ranking reduces cognitive load, or that an artist would benefit from seeing all five rather than a differently organized field.

Future validation should therefore extend in at least three directions. First, professional artists and curators should compare proposal sets and individual interventions to test construct validity of the model-mediated preferences. Second, longitudinal interaction studies should test whether multiple ranked alternatives, editable search states, and recoverable branches actually support exploration rather than fixation or cognitive overload. Third, selected proposals should be followed through technical review and, eventually, fabrication and situated encounter. These studies answer questions that no larger model-judge benchmark can substitute for.

11 Conclusion

A consequential part of contemporary-art practice occurs before an artwork has stabilized: an artist may know what problem, pressure, site, material condition, or institutional constraint matters without yet knowing what the work itself should become. This paper has treated that interval as an explicit computational task. AIrtist develops an under-specified artist-authored installation brief into several complete project hypotheses and models development as contestable transformation: proposed changes must compete with the artistic state they are intended to replace rather than being accepted simply because additional inference has produced them.

The empirical results support this formulation, but not uniformly across all stages. AIrtist produces a stronger measured five-proposal set than all three planned controls, including the substantially matched M5 condition that starts from the same five self-refined parents and closely matches the iterative scaffold. Against M5, the centered complete-pool effect is +0.0836 (95% CI [+0.0340, +0.1304], Holm $p = .0015$). The gain is therefore difficult to explain as a consequence of better starting samples or merely generating more generic continuations from the same parents. At the intervention level, 59.6% of final lane states remain content-identical to their incumbents; among the 101 replacements actually adopted, challengers receive mean external preference 0.693 against the exact states they displace, with a brief-balanced centered effect of +0.437. These results do not identify which individual reviewer, opportunity-state field, or intervention role is causally necessary, and they do not establish that incumbent retention is superior to forced replacement. They do show that domain-structured development can improve the proposal field while preserving non-intervention as a legitimate outcome.

Terminal selection is less resolved. The Hybrid selector has favorable point estimates relative to Base, but its +0.0533 Utility@1 advantage is not statistically resolved. Stronger candidates often already exist inside the generated five-proposal pool without being surfaced first. The experiment therefore establishes the recognition problem more clearly than it establishes the present solution: producing promising material and recognizing which existing candidate deserves attention are empirically separable capabilities.

This separation is the broader methodological lesson of the study. A weak surfaced proposal can arise because the system failed to construct strong alternatives, because an intervention damaged a stronger state, or because ranking hid useful material that had already been generated. Evaluating only one final answer collapses these failure modes. AIrtist instead makes candidate-set quality, intervention quality, and attention allocation separately measurable. The post-hoc breadth analysis adds a related caution: the higher measured quality of A5 relative to M5 did not coincide with greater measured breadth under the chosen latent-semantic representation. Better proposal sets and broader exploration should therefore be treated as distinct objectives rather than assuming that more agents, branches, or outputs automatically create a wider creative field.

Contemporary-art installation is central to this argument rather than merely a difficult benchmark. In this domain, a change in material causality, spatial organization, spectator position, or institutional procedure can alter the identity of a proposal rather than simply improve its presentation. A critique can correctly diagnose a weakness while the proposed repair destroys the strongest relation in the parent. This is why preservation, transformation, and selection need to remain visible as separate decisions. Similar problems arise in other open-ended creative domains, but installation practice makes their consequences unusually explicit.

The claims of the present study remain bounded. The evaluated objects are textual project hypotheses rather than realized artworks; the principal quality signal is model-mediated; and no artist negotiates with the system inside the frozen experimental pipeline. The results therefore concern a computational development policy under a disclosed working quality model, not professional curatorial consensus, cultural value, or demonstrated improvement to human creativity. The next step is to expose the same development process to negotiated co-creativity, so that artists can contest diagnoses, preservation constraints, candidate priorities, and discarded branches rather than receiving only a finished ranking.

The aim is not to automate the decision of what the artwork should be. It is to make the space in which that decision is formed richer, more inspectable, and less prematurely collapsed.

References

[1] Claire Bishop. *Installation Art: A Critical History*. London: Tate Publishing, 2005.

[2] Miwon Kwon. *One Place After Another: Site-Specific Art and Locational Identity*. Cambridge, MA: MIT Press, 2002.

[3] Muzeum Sztuki in Łódź. *Errorism: Agnieszka Kurant*. Exhibition brochure. 2021. URL: `https://msl.org.pl/sites/msl/files/aktualnosci/Errorism-accessible-brochure.pdf` (visited on 08/28/2026).

[4] Jonas Lund. *In the Middle of Nowhere*. Artist project documentation. 2023. URL: `https://jonaslund.com/works/in-the-middle-of-nowhere/` (visited on 08/28/2026).

[5] Cai Guo-Qiang Studio. *cAI: An Artificial Intelligence Project*. 2023. URL: `https://caiguoqiang.com/projects/projects-2023/` (visited on 08/28/2026).

[6] J. Paul Getty Trust. *PST ART: Art & Science Collide to Kick-Off with Fireworks by Artist Cai Guo-Qiang*. 2024. URL: `https://www.getty.edu/news/pst-art-art-science-collide-to-kick-off-with-fireworks-by-artist-cai-guo-qiang/` (visited on 08/28/2026).

[7] Wen-Fan Wang, Chien-Ting Lu, Nil Ponsa i Campanyà, Bing-Yu Chen, and Mike Y. Chen. "AIdeation: Designing a Human-AI Collaborative Ideation System for Concept Designers". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Article 21. Association for Computing Machinery, 2025, pp. 1–28. DOI: 10.1145/3706598.3714148.

[8] Jinheon Baek, Sujay Kumar Jauhar, Silviu Cucerzan, and Sung Ju Hwang. "ResearchAgent: Iterative Research Idea Generation over Scientific Literature with Large Language Models". In: *Proceedings of NAACL 2025*. Association for Computational Linguistics, 2025, pp. 6709–6738. DOI: 10.18653/v1/2025.naacl-long.342. URL: https://aclanthology.org/2025.naacl-long.342/ (visited on 08/28/2026).

[9] Juraj Gottweis et al. *Towards an AI Co-Scientist*. 2025. arXiv: 2502.18864. URL: https://arxiv.org/abs/2502.18864 (visited on 08/28/2026).

[10] Samangi Wadinambiarachchi, Ryan M. Kelly, Saumya Pareek, Qiushi Zhou, and Eduardo Velloso. "The Effects of Generative AI on Design Fixation and Divergent Thinking". In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. Article 380. Association for Computing Machinery, 2024, pp. 1–18. DOI: 10.1145/3613904.3642919.

[11] Anil R. Doshi and Oliver P. Hauser. "Generative AI Enhances Individual Creativity but Reduces the Collective Diversity of Novel Content". In: *Science Advances* 10.28 (2024), eadn5290. DOI: 10.1126/sciadv.adn5290.

[12] Anne Ring Petersen. *Installation Art between Image and Stage*. Copenhagen: Museum Tusculanum Press, 2015.

[13] Brian O'Doherty. *Inside the White Cube: The Ideology of the Gallery Space*. Santa Monica: Lapis Press, 1986.

[14] Juliane Rebentisch. *Aesthetics of Installation Art*. Berlin: Sternberg Press, 2012.

[15] Robert Morris. "Notes on Sculpture, Part 2". In: *Artforum* 5.2 (1966), pp. 20–23.

[16] Jack Burnham. "Systems Esthetics". In: *Artforum* 7.1 (1968), pp. 30–35.

[17] Museum of Modern Art. *James Turrell. Frontal Passage. 1994*. Collection record; artwork date 1994. URL: https://www.moma.org/collection/works/81089 (visited on 08/28/2026).

[18] Museum of Modern Art. *Mel Bochner. Measurement Room. 1969*. Collection record; artwork date 1969. URL: https://www.moma.org/collection/works/79441 (visited on 08/28/2026).

[19] MACBA – Museu d'Art Contemporani de Barcelona. *Hans Haacke, Condensation Cube*. Collection record; artwork date 1963–1967. URL: https://www.macba.cat/en/obra/r1523-condensation-cube/ (visited on 08/28/2026).

[20] Doris Salcedo. *Shibboleth, 2007: Artist Statement*. Signs: Journal of Women in Culture and Society online feature; artwork date 2007. 2019. URL: https://signsjournal.org/doris-salcedo-shibboleth-2007/ (visited on 08/28/2026).

[21] Art Institute of Chicago. *Felix Gonzalez-Torres, Untitled (Portrait of Ross in L.A.)* Collection educational documentation; artwork date 1991. URL: https://www.artic.edu/assets/f85bee6b-d18b-5577-a9e6-f429421b4654 (visited on 08/28/2026).

[22] Tate. *Tania Bruguera – Tatlin's Whisper #5*. TateShots video documentation; artwork date 2008. 2008. URL: https://www.youtube.com/watch?v=x7L1s_GWn3o (visited on 08/28/2026).

[23] Museum of Modern Art. *Rirkrit Tiravanija, untitled (free), 1992*. Collection record / institutional documentation; work originated in 1992. URL: https://www.moma.org/collection/works/147206 (visited on 08/28/2026).

[24] Art21. *Fred Wilson: Mining the Museum, 1992–1993*. Institutional educational documentation. URL: https://art21.org/gallery/fred-wilson-artwork-survey-1990s/ (visited on 08/28/2026).

[25] Jacques Rancière. *The Emancipated Spectator*. London: Verso, 2009.

[26] Claire Bishop. *Artificial Hells: Participatory Art and the Politics of Spectatorship*. London: Verso, 2012.

[27] Giuliana Bruno. *Surface: Matters of Aesthetics, Materiality, and Media*. Chicago: University of Chicago Press, 2014.

[28] Graeme Ritchie. “Some Empirical Criteria for Attributing Creativity to a Computer Program”. In: *Minds and Machines* 17.1 (2007), pp. 67–99.

[29] Anna Jordanous. “A Standardised Procedure for Evaluating Creative Systems: Computational Creativity Evaluation Based on What It Is to Be Creative”. In: *Cognitive Computation* 4 (2012), pp. 246–279.

[30] Pierre Bourdieu. *The Field of Cultural Production: Essays on Art and Literature*. New York: Columbia University Press, 1993.

[31] Arthur C. Danto. “The Artworld”. In: *The Journal of Philosophy* 61.19 (1964), pp. 571–584.

[32] Umberto Eco. *The Open Work*. Cambridge, MA: Harvard University Press, 1989.

[33] Sianne Ngai. *Theory of the Gimmick: Aesthetic Judgment and Capitalist Form*. Cambridge, MA: Belknap Press of Harvard University Press, 2020.

[34] Howard S. Becker. *Art Worlds*. Berkeley: University of California Press, 1982.

[35] Shannon Jackson. *Social Works: Performing Art, Supporting Publics*. New York: Routledge, 2011.

[36] Grant H. Kester. *Conversation Pieces: Community and Communication in Modern Art*. Berkeley: University of California Press, 2004.

[37] Hal Foster. *The Return of the Real: The Avant-Garde at the End of the Century*. Cambridge, MA: MIT Press, 1996.

[38] William W. Gaver, Jacob Beaver, and Steve Benford. “Ambiguity as a Resource for Design”. In: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2003, pp. 233–240.

[39] Kristina Höök, Phoebe Sengers, and Gerd Andersson. “Sense and Sensibility: Evaluation and Interactive Art”. In: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2003, pp. 241–248.

[40] Margaret A. Boden. “Creativity and Artificial Intelligence”. In: *Artificial Intelligence* 103.1–2 (1998), pp. 347–356.

[41] Margaret A. Boden. *The Creative Mind: Myths and Mechanisms*. 2nd ed. London: Routledge, 2004.

[42] Geraint A. Wiggins. “A Preliminary Framework for Description, Analysis and Comparison of Creative Systems”. In: *Knowledge-Based Systems* 19.7 (2006), pp. 449–458.

[43] David G. Jansson and Steven M. Smith. “Design Fixation”. In: *Design Studies* 12.1 (1991), pp. 3–11.

[44] Gazelli Art House. *I'm Not A Painter: Alexander Reben*. Exhibition documentation. 2022. URL: `https://gazelliarthouse.com/exhibitions/158-im-not-a-painter-alexander-reben/` (visited on 08/28/2026).

[45] Anicka Yi Studio. *There Exists Another Evolution, But In This One*. Leeum Museum of Art. 2024. URL: `https://www.anickayistudio.biz/exhibitions/leeum` (visited on 08/28/2026).

[46] Serpentine Galleries. *Holly Herndon and Mat Dryhurst: The Call*. Exhibition, 2024–2025. 2024. URL: `https://www.serpentinegalleries.org/about/press/holly-herndon-and-mat-dryhurst-the-call/` (visited on 08/28/2026).

[47] Thomas F. Eisenmann, Andres Karjus, Mar Canet Sola, Levin Brinkmann, Bramantyo Ibrahim Supriyatno, and Iyad Rahwan. “Expertise Elevates AI Usage: Experimental Evidence Comparing Laypeople and Professional Artists”. In: *International Journal of Human–Computer Interaction* (2026). DOI: `10.1080/10447318.2026.2669041`.

[48] Sonja Rozental and Alwin de Rooij. “Neither Tool nor Collaborator: Rethinking Human–AI Co-Creativity in Artistic Practice with Material Engagement Theory”. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. Article 116. Association for Computing Machinery, 2026, pp. 1–20. DOI: `10.1145/3772318.3791752`.

[49] Shuai Liu, Yiqing Tian, Yang Chen, and Mar Canet Sola. *Artism: AI-Driven Dual-Engine System for Art Generation and Critique*. NeurIPS 2025 Creative AI Track. 2025. URL: `https://proceedings.neurips.cc/paper_files/paper/2025/hash/3d2c2f19aa4ca9493813f13aa044f926-Abstract-Creative_AI_Track.html` (visited on 08/28/2026).

[50] Pei Chen, Jiayi Yao, Zhuoyi Cheng, Yichen Cai, Jiayang Li, Weitao You, and Lingyun Sun. "CoExploreDS: Framing and Advancing Collaborative Design Space Exploration Between Human and AI". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Article 22. Association for Computing Machinery, 2025, pp. 1–20. DOI: 10.1145/3706598.3713869.

[51] Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. "Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers". In: *International Conference on Learning Representations*. 2025. URL: https://proceedings.iclr.cc/paper_files/paper/2025/hash/ea94957d81b1c1caf87ef5319fa6b467-Abstract-Conference.html (visited on 08/28/2026).

[52] Barrett R. Anderson, Jash Hemant Shah, and Max Kreminski. "Homogenization Effects of Large Language Models on Human Creative Ideation". In: *Proceedings of the 16th Conference on Creativity & Cognition*. 2024, pp. 413–425. DOI: 10.1145/3635636.3656204.

[53] Nuo Chen, Yicheng Tong, Yuzhe Yang, Yufei He, Xueyi Zhang, Zou Qingyun, Qian Wang, and Bingsheng He. "Diversity Collapse in Multi-Agent LLM Systems: Structural Coupling and Collective Failure in Open-Ended Idea Generation". In: *Findings of the Association for Computational Linguistics: ACL 2026*. San Diego, California, United States: Association for Computational Linguistics, 2026, pp. 251–306. DOI: 10.18653/v1/2026.findings-acl.13. URL: https://aclanthology.org/2026.findings-acl.13/.

[54] Alejandro Hernandez Artiles, Hiromu Yakura, Levin Brinkmann, Mar Canet Sola, Hassan Alhaija, Ignacio Serna, Nasim Rahaman, Bernhard Schölkopf, and Iyad Rahwan. "Cultural Alien Sampler: Open-Ended Art Generation Balancing Originality and Coherence". In: *Advances in Neural Information Processing Systems 38 Creative AI Pre-Proceedings*. Vol. 38. 2025. URL: https://proceedings.neurips.cc/paper_files/paper/2025/hash/ce57a4b9274d44d7692e715ed1876ded-Abstract-Creative_AI_Track.html (visited on 08/28/2026).

[55] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. "Self-Refine: Iterative Refinement with Self-Feedback". In: *Advances in Neural Information Processing Systems*. Vol. 36. 2023.

[56] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. "Tree of Thoughts: Deliberate Problem Solving with Large Language Models". In: *Advances in Neural Information Processing Systems*. Vol. 36. 2023.

[57] Kavana Venkatesh, Connor Dunlop, and Pinar Yanardag. "CREA: A Collaborative Multi-Agent Framework for Creative Image Editing and Generation". In: *Advances in Neural Information Processing Systems*. Vol. 38. 2025. URL: https://proceedings.neurips.cc/paper_files/paper/2025/hash/fa41e9d5dfcc97cd9eed99f001aa28e5-Abstract-Conference.html (visited on 08/28/2026).